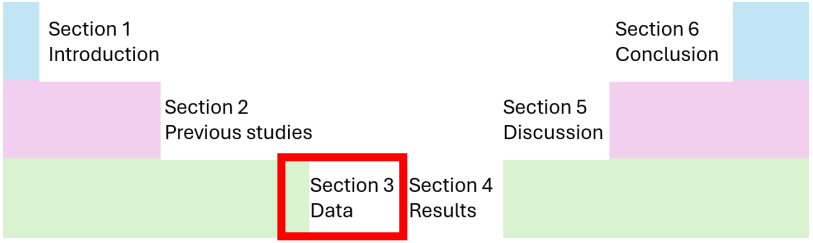
Class 5

Academic reading 3: Data & Results

Corpus research 2: UD (part 1)



In this section, I examine the issues raised above on the previously suggested accounts of the 'passive' unaccusative structure in light of further data obtained from a large database of non-native English.

1 Data

The data I discuss below comes from the Longman Learners Corpus (Version 1.1., March 1993, henceforth, LLC), which is a large computerized database of primarily written English produced by native speakers of various L1s. Data from native speakers of Italian, Spanish, Japanese and Korean were chosen for the study (primarily due to the availability of a relatively large sample size) and a total of 3362 essays were separated as a research corpus from the entire LLC. The number of essays in each L1 group was 684 for Italian, 1079 for Spanish, 236 for Korean, and 1363 for Japanese. Although the LLC encodes variables such as the learner's proficiency level, the environment in which the data were produced (e.g., examination, classwork, homework, business, etc.), the task type (e.g., essay, report, diary, etc.), the target language variety (British, American or Australian English) and so forth, only L1 was taken into account in this study.

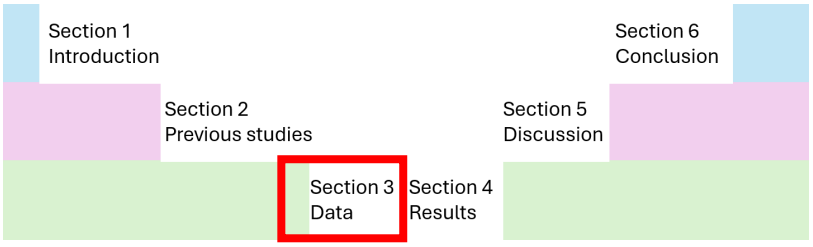
Section 3 The corpus analysis

Data

Longman Learners Corpus

Speaker: Italian, Spanish, Japanese, and Korean

Essays: 3,362



Only nonalternating unaccusative verbs were chosen for the study so that any instance of 'passive' or 'causative' use could be unmistakably identified as an IL innovation. Ten verbs were preselected because of their common appearance in the 'passive' form in IL English, which has been widely documented in the literature.

32) *Nonalternating unaccusatives*
appear arise arrive die[11] disappear
exist fall happen occur rise

Sentences with an unaccusative verb were extracted from the research corpus by a search program together with the lines immediately preceding and following them. The additional lines were taken into consideration when a token sentence alone did not provide enough information to interpret the use of the verb intended by the learner. When further information was needed, the original essay was consulted as well.

The token sentences were classified into the 10 structural patterns listed below. Eight of these (33a)–(33h), were logical possibilities based on the following criteria:

1) the position of the NP argument vis-à-vis the verb (i.e., preverbal or postverbal);
2) the type of verbal phrase (i.e., the existence or lack of the *be* + *en*); and
3) the existence and type of expletive subject (i.e., *there*, *it* or *pro*).

Section 3 The corpus analysis

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Longman Learners Corpus

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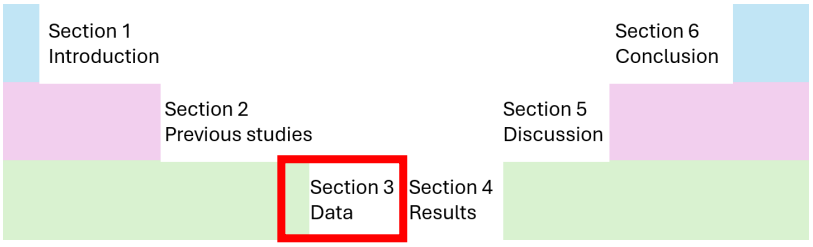
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1) Verbs: appear, arise, arrive, die, disappear, exist, fall, happen, occur, rise
2) Classification:

10 structural patterns

a. N < V vs. V < N
b. V vs. be V-en
c. there vs. N



In order to accommodate prominent structural patterns that did not fall into any of the eight patterns, the causative structure of (33i) and the existential pseudo-relative structure of (33j) (Jordens, 1995; Yip, 1995: Chapter 7) were added. The ten patterns are listed below with hypothetical examples[12]

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Section 3 The corpus analysis

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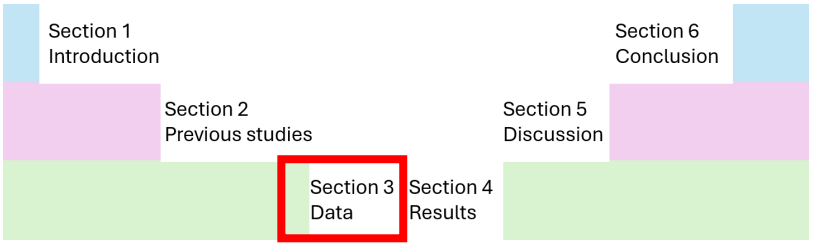
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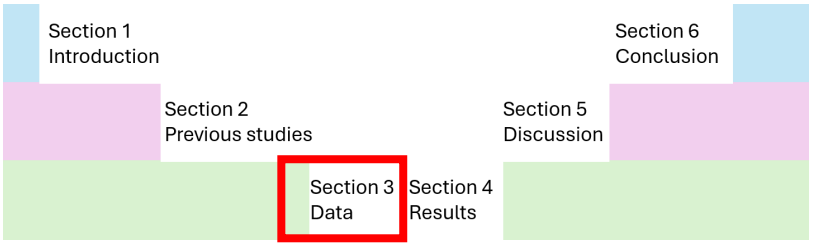
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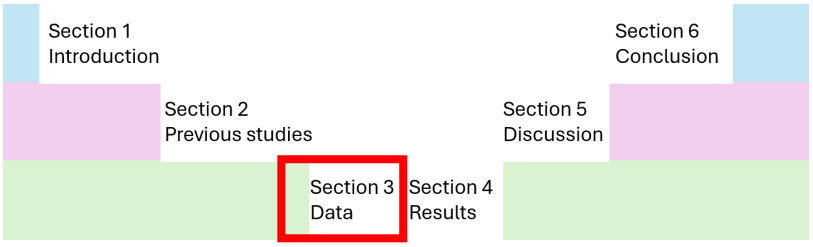
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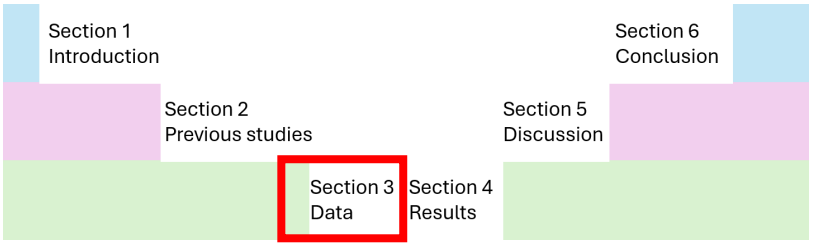
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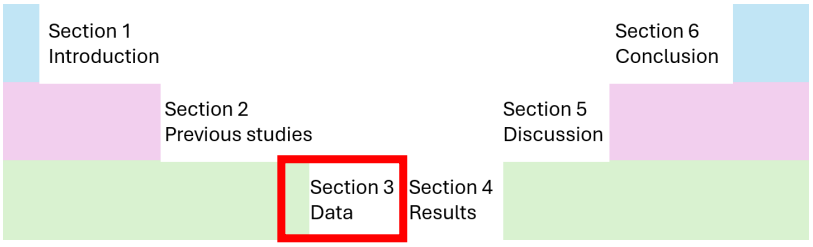
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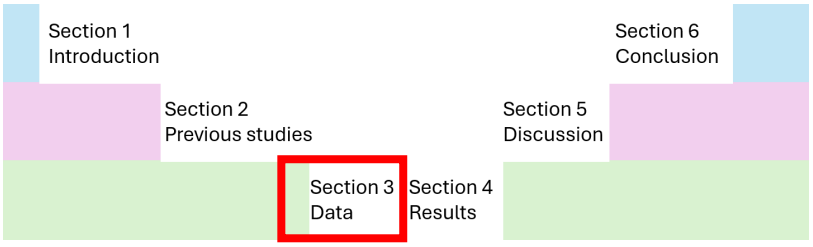
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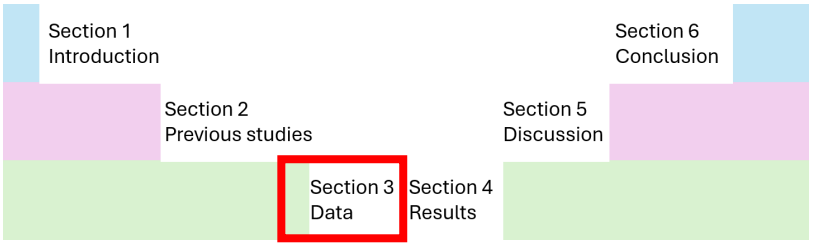
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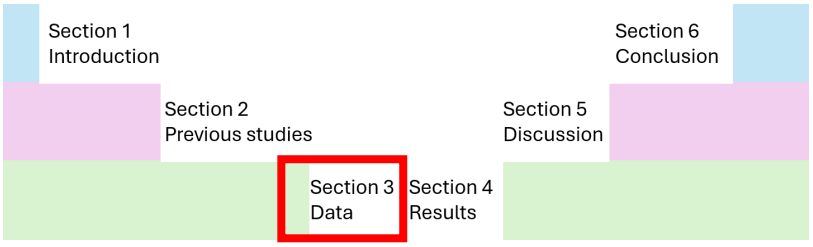
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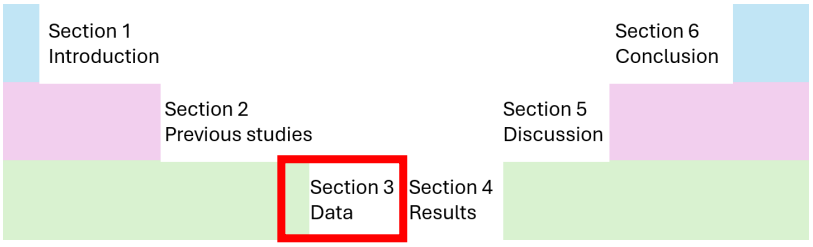
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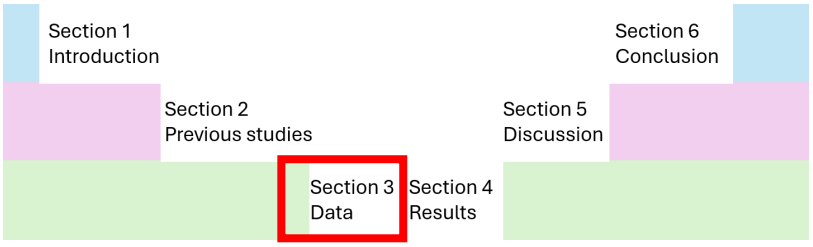
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Among the sentences obtained from the research corpus, some types of verb forms and usage were excluded from the final tally.

34) *Verb forms and usages excluded from the final tally*
- a. unaccusative verbs with propositional complements, namely, raising verbs; e.g., *appear (to be) happy*, *happen to be in the room*, etc.
- b. idioms and metaphorical usage of verbs; e.g., *fall in love*, *fall ill*, etc.
- c. nonfinite verbs; e.g., infinitives (with or without *to*), gerunds, and participle constructions

Raising verbs (*appear* and *happen*) were excluded because they take a clausal, not nominal, complement. Idioms and metaphorical use of verbs were also screened out because they tend to appear repeatedly within a single essay and may thus skew the overall result. Nonfinite verbs were ignored because it is impossible to determine the syntactic position of a covert argument (PRO) relative to the verb. The preselection of verbs and the exclusion of certain types of tokens contrast with other corpus-based studies in which all unaccusative-looking tokens were gathered (e.g., Zobl, 1989; Han, 1998). Thanks to this methodological rigour, the determination and collection of 'passive' unaccusatives was much less arbitrary than in previous studies, although it no doubt contributed to the relatively low percentage of tokens as we see later.

Finally, the 10 unergative verbs in (35) were also preselected and the syntactic distribution of their tokens was contrasted with that of the unaccusatives:

35) *Unergative verbs*

cough	cry	dance	joke	laugh
shout	smile	speak	talk	walk

Section 3 The corpus analysis

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Longman Learners Corpus

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3) Excluded forms

a. Raising predicates

b. idioms

c. Non-finite uses

4) Reference group: unergatives

cough, cry, dance, joke, laugh, shout, smile, speak, talk, walk



A total of 941 token sentences were obtained on the unaccusative verbs: 240 for L1 Italian, 347 for L1 Spanish, 50 for L1 Korean and 304 for L1 Japanese. Table 1 shows the overall syntactic distribution of the unaccusative verbs by the L1 of the subjects (see Oshita, 1997: Appendixes A and B for more information).

Despite the enormous differences among the L1 groups concerning the frequency of 'passive' unaccusative tokens, Table 1 clearly shows that among the non NP–V structures, the NP–*be*–V*en* structure was the pattern most commonly used by all the L1 groups combined, confirming the previous observations of its appearance in IL English.[13]

The search for the 10 preselected unergative verbs in the same research corpus produced 640 token sentences. Among them, only one instance of a 'passive' structure was found; the example was *He has been walked since last month* produced by a L1 Spanish speaker. This finding is quite significant by itself because it supports Hubbard's (1983 reported in Hubbard, 1994: 55–56) claim that virtually all the instances of nontarget 'passive' structures on intransitive verbs involve unaccusatives, not unergatives.

Section 3 The corpus analysis

Results

1) Unaccusatives: 38 instances

2) Unergatives: 1 instance

Table 1 Overall syntactic distribution of unaccusative verbs

Structural patterns	Italian	Spanish	Korean	Japanese	Total
a. NP–V	216	327	39	269	851
b. NP–*be*–V*en*	8	5	8	17	38
c. *there*–V–NP	2	0	1	1	4
d. *it*–V–NP	5	6	1	1	13
e. ø–V–NP	7	8	0	1	16
f. *there*–*be*–V*en*–NP	0	0	0	4	4
g. *it*–*be*–V*en*–NP	0	0	0	0	0
h. ø–*be*–V*en*–NP	0	0	0	0	0
i. NP1–V–NP2	2	1	1	7	11
j. *there*–*be*–NP–V	0	0	0	4	4
Total	240	347	50	304	941

Task 3 Single word search

https://keiosfc-aki2025.github.io/KeioSFC/keioSFC.html



Task 3 Search an expression using KeioSFC

100件ヒットしました

☞ 検索結果

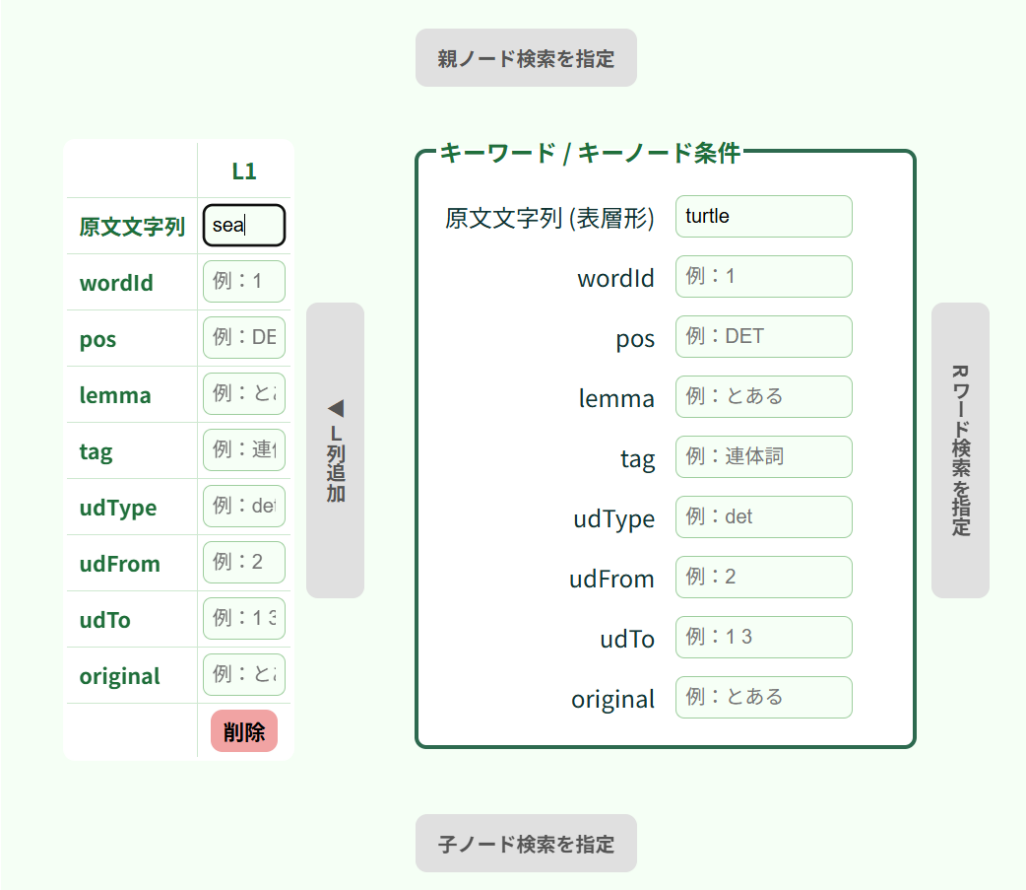
1 / 10 ページ (全 100 件) 次の10件を表示

通し番号	sentID	タスク番号	作文番号	個人ID	年齢	使用言語	第一言語	第二言語	前文脈	キー	後文脈
1	1	1	19	14	18	English	日	English	Once upon a time , the small Onceuponatime,thesmall	turtle turtle	was in a bottle . wasinabottle.
2	4	1	19	14	18	English	日	English	He was trying to let escape the Hewastryingtoletescapethe	turtle turtle	from the bottle , bu . fromthebottle,buthe
3	5	1	19	14	18	English	日	English	The The	turtle turtle	was grown twenty t wasgrowntwentytime
									The	turtle	also went to there

Task 4

Search a sequence

https://keiosfc-aki2025.github.io/KeioSFC/keioSFC.html



Task 4 Search a sequence

前文脈	キー	後文脈
There is a bottle floating on the ocean in which a small (sea) Thereisabottlefloatingontheoceaninwhichasmall(sea)	turtle turtle	is contained . iscontained.
However , upon touching , the bottle exploded and the captured (sea) However,upontouching,thebottleexplodedandthecaptured(sea)	turtle turtle	has jumped out . hasjumpedout.
It becomes larger and larger , and it turns into a mature (sea) Itbecomeslargerandlarger,anditturnsintoamature(sea)	turtle turtle	. .
It is actually not letters , but picutres that are drawn on the paper : a handsome young man , an a treasure box , and the (sea) Itisactuallynotletters,butpicutresthataredrawnonthepaper:ahandsomeyoungman,anatreasurebox,andthe(sea)	turtle turtle	in the bottle that we saw earl inthebottlethatwesawearlierinth
Just as the bottle broke down when he tought it in his dream , the treasure box becomes shiny and breaks down , and a big (sea)	turtle	comes out .

Task 5 Dependencies

Goal: Obtain passivized unaccusatives

The boys were arrived.

(1) First shot:
L1 word: [lemma: be & tag: VBZ]
Key: [tag: VBN]

The boys were **almost** arrived.

Task 5 Dependencies

Goal: Obtain passivized unaccusatives

The boys were arrived.

The boys were **almost** arrived.

(2) Second shot:

L1 word: [lemma: be & tag: VBZ]
Key: [tag: VBN]

L2 word: [lemma: be & tag: VBZ]
Key: [tag: VBN]

L3 word: [lemma: be & tag: VBZ]
Key: [tag: VBN]

Task 5

Dependencies

Goal: Obtain passivized unaccusatives

The boys **were arrived**.

The boys **were <u>almost</u> arrived**.

The boys **were** here who **arrived** early.

Universal Dependencies

A visual guide to all 37 universal relation labels (UD v2), with tree diagrams for every example



What Is Universal Dependencies?

One shared way to describe the grammar of any language

Universal Dependencies (UD) is an open, community-built framework for annotating grammar consistently across human languages. A corpus annotated this way is called a **treebank**. The UD project lists over 200 treebanks in over 150 languages.

UPOS

Part of speech

17 universal word classes such as NOUN, VERB and ADJ.

```
dogs → NOUN
```

FEATS

Morphological features

Grammatical categories such as number, tense and case.

```
dogs → Number=Plur
```

DEPREL

Dependency relations

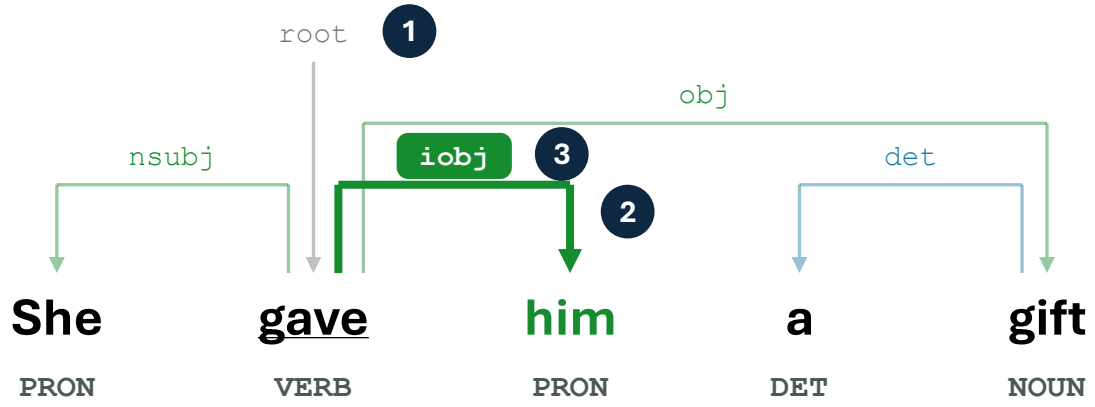
37 universal labels for how a word relates to its head. This deck is about these.

```
dogs ← bark : nsubj
```

What is a dependency tree? It has one node per word and no phrase nodes such as NP or VP. Every word depends on exactly one head word; only the root has none. The arrows between words, and their labels, are what this deck explains.

How to Read the Arrows

A worked example: “She gave him a gift”



1 Find the root

The main predicate, “gave”, has no head. The only arrow into it comes from the root marker.

2 Follow one arrow

The arrow runs from the head “gave” to the word that depends on it, “him”. The arrowhead marks the dependent.

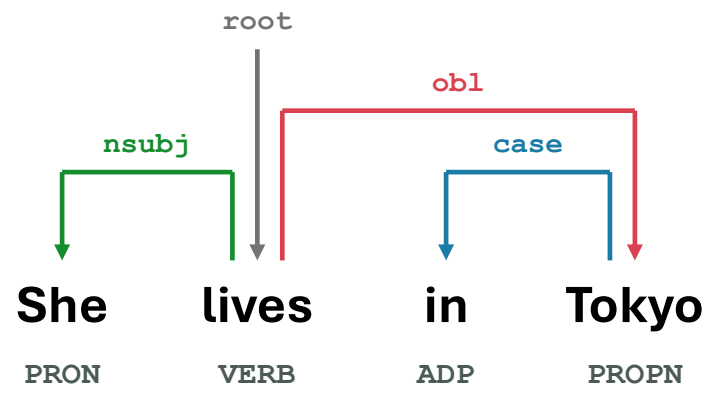
3 Read the label

The label tells how the dependent relates to its head: “him” is the recipient, so iobj. Every word except the root has exactly one head.

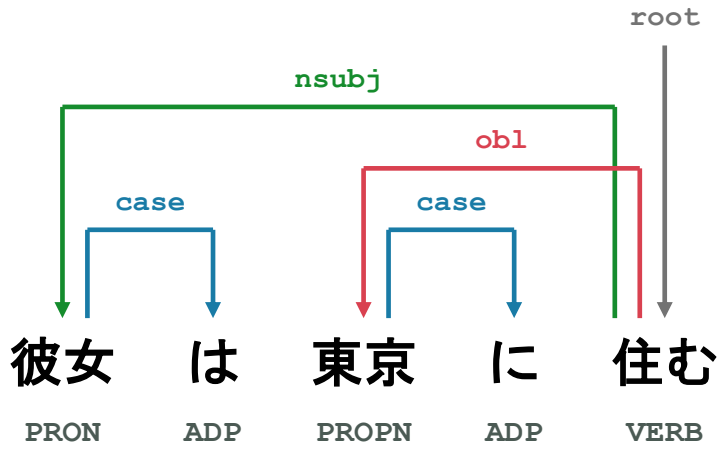
One Scheme for Many Languages

Same labels, different word order: why UD is built this way

English · head-initial



日本語 · head-final



‘She lives in Tokyo.’

Design intent

One construction gets one label in every language. Content words are the heads; function words (case, aux, mark) attach to them, so structures line up across languages.

Other languages

Word order, particles and morphology differ, but the labels stay. “in Tokyo” and 東京に both give obl + case; only the order and arrow direction change.

Why it helps

Treebanks can be compared and pooled, parsers can transfer from one language to another, and linguists can measure differences between languages with one set of labels.

Relation Families at a Glance

37 universal relations in 8 families (UD v2)

Core arguments

6 relations

`nsubj` `obj` `iobj` `csubj` `ccomp` `xcomp`

Non-core dependents

10 relations

`obl` `vocative` `expl` `dislocated` `advcl` `advmod` `discourse` `aux` `cop` `mark`

Nominal dependents

8 relations

`nmod` `appos` `nummod` `acl` `amod` `det` `clf` `case`

Coordination

2 relations

`conj` `cc`

Headless

2 relations

`fixed` `flat`

Loose

2 relations

`list` `parataxis`

Special

4 relations

`compound` `orphan` `goeswith` `reparandum`

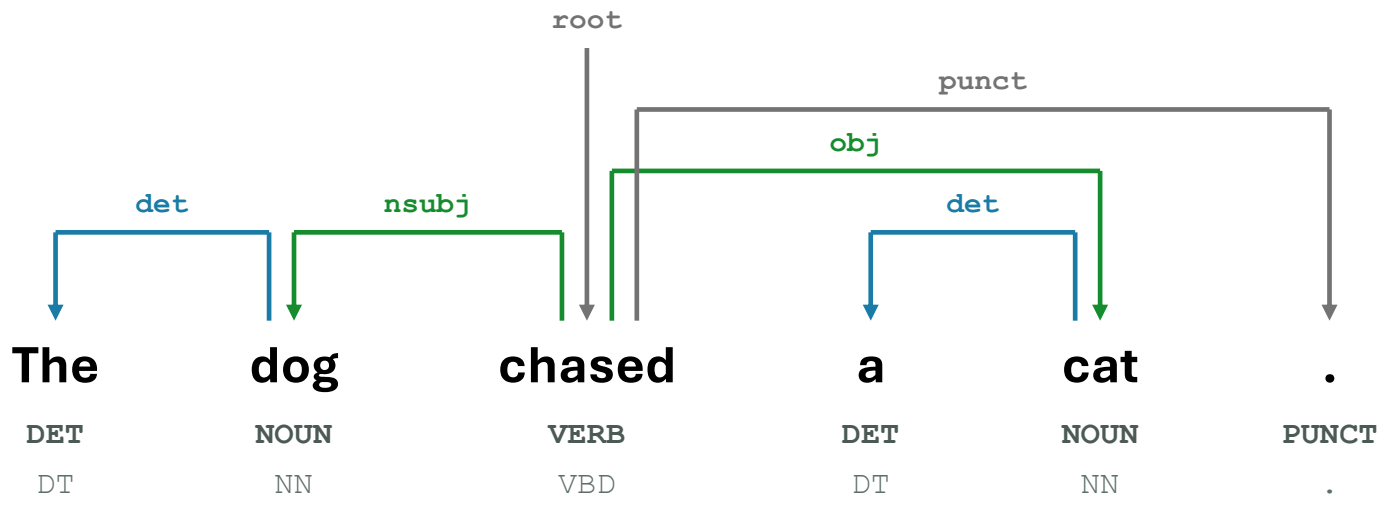
Other

3 relations

`punct` `root` `dep`

Reading a Dependency Tree

Arcs show head → dependent; labels are relations; under each word: UPOS, then Penn Treebank tag



UPOS · Penn Treebank

Head → dependent

Each arrow runs from a head word to the word that depends on it. The root has no head.

Relation on the arc

The label above each arc names the relation, e.g. det, nsubj, obj. All 37 are covered next.

UPOS and PTB below

Under every word: its Universal POS tag (DET, NOUN, VERB) and, below it, the Penn Treebank tag.

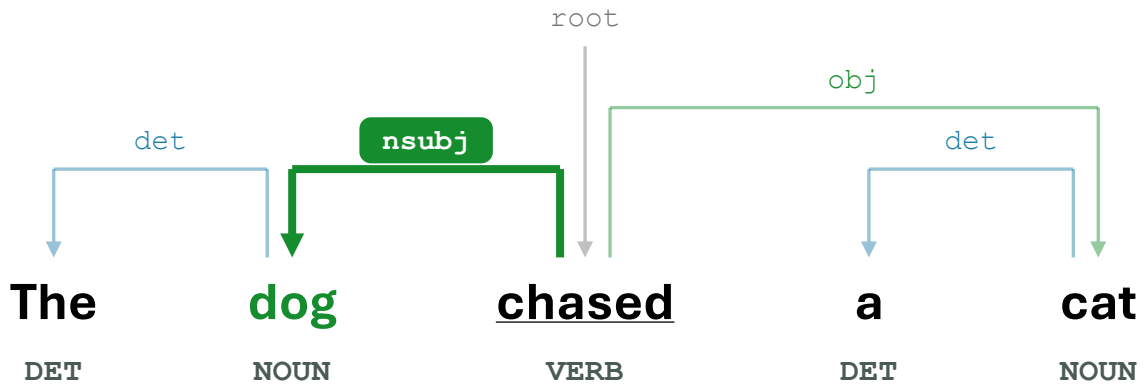
UPOS Tags Used as Gloss

The 17 Universal part-of-speech tags that appear under the words in every tree

ADJ **adjective** big, old, green	**ADP** **adposition** in, to, during	**ADV** **adverb** very, well, exactly
AUX **auxiliary** is, has, will	**CCONJ** **coordinating conjunction** and, or, but	**DET** **determiner** a, the, this
INTJ **interjection** psst, ouch, bravo	**NOUN** **noun** girl, cat, tree	**NUM** **numeral** 1, one, seventy
PART **particle** ’s, not, to (infinitive)	**PRON** **pronoun** I, you, who	**PROPN** **proper noun** Mary, NATO
PUNCT **punctuation** . , (	**SCONJ** **subordinating conjunction** if, while, that	**SYM** **symbol** $, %, +
VERB **verb** run, eat, chase	**X** **other** foreign words, typos	**17 tags**

`nsubj` Nominal subject

Subject of a clause: the noun that performs the action or is described.



EXAMPLE · DEPENDENT ← HEAD

dog ← chased

“The dog chased a cat”

CORE ARGUMENTS

`nsubj` `obj` `iobj` `csubj` `ccomp` `xcomp`

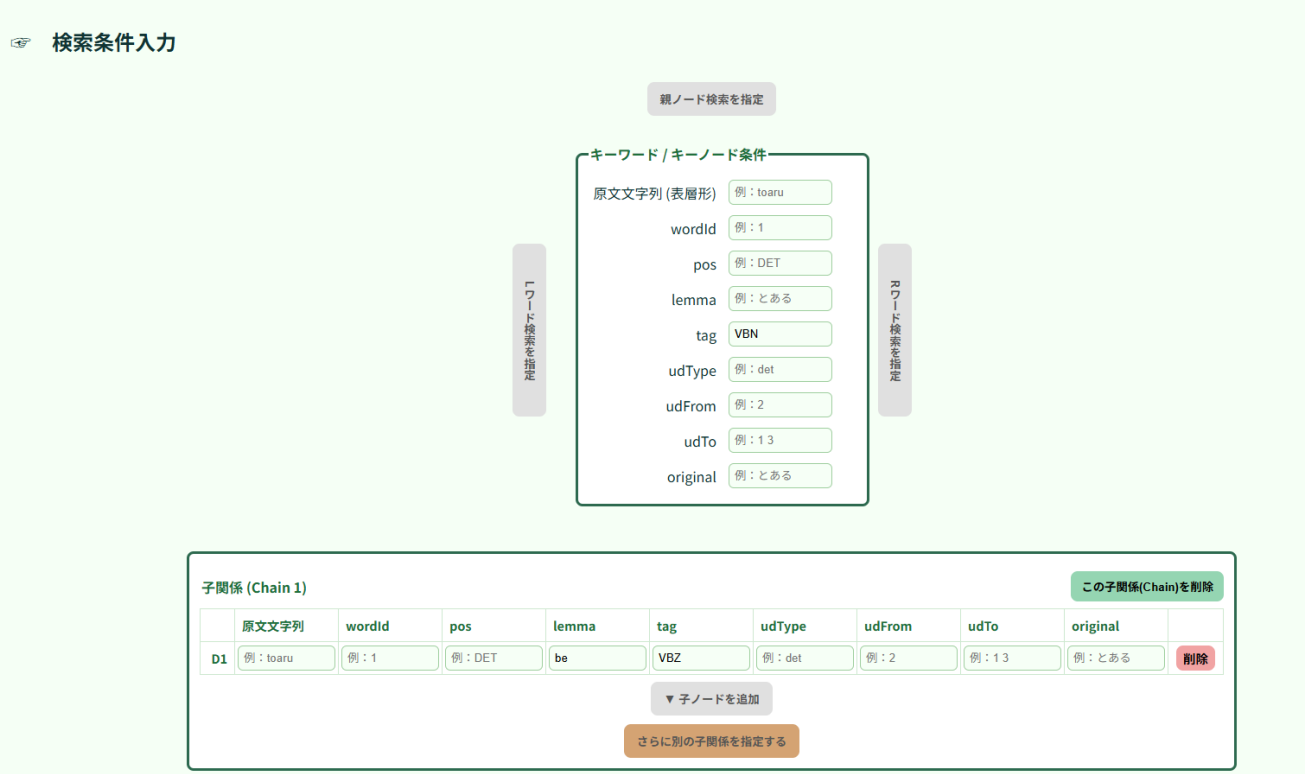
Task 5 Dependencies

Goal: Obtain passivized unaccusatives

The boys **were arrived**.

The boys **were almost arrived**.

The boys **were** here who **arrived** early.



Task 5

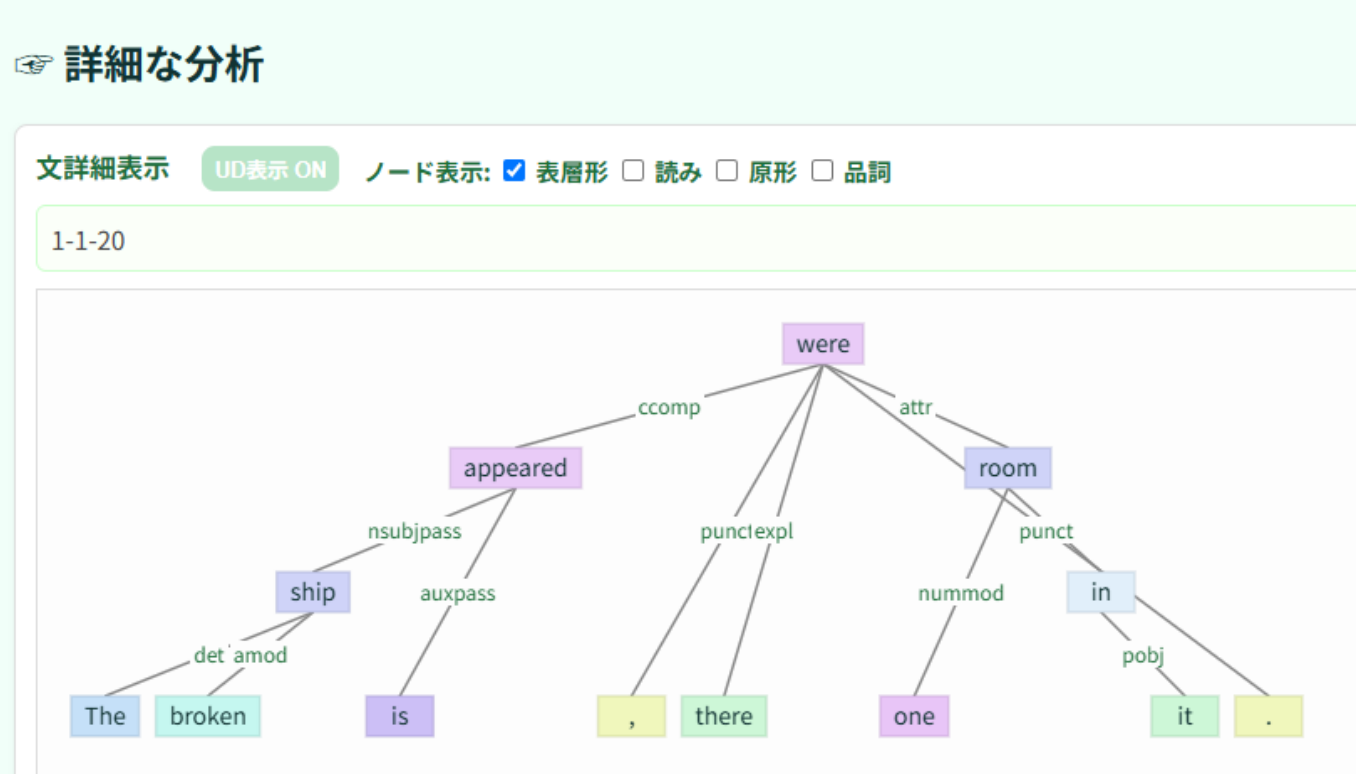
Dependencies

Goal: Obtain passivized unaccusatives

The boys **were arrived**.

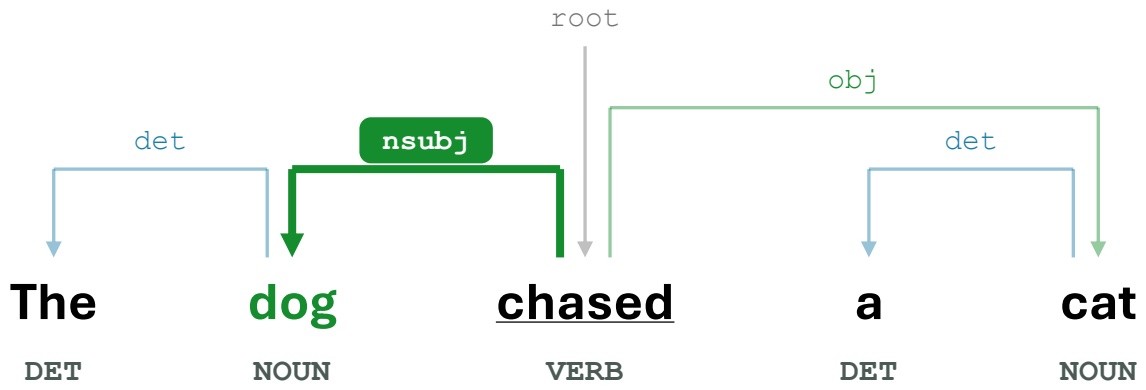
The boys **were almost arrived**.

The boys **were** here who **arrived** early.



`nsubj` Nominal subject

Subject of a clause: the noun that performs the action or is described.



EXAMPLE · DEPENDENT ← HEAD

dog ← chased

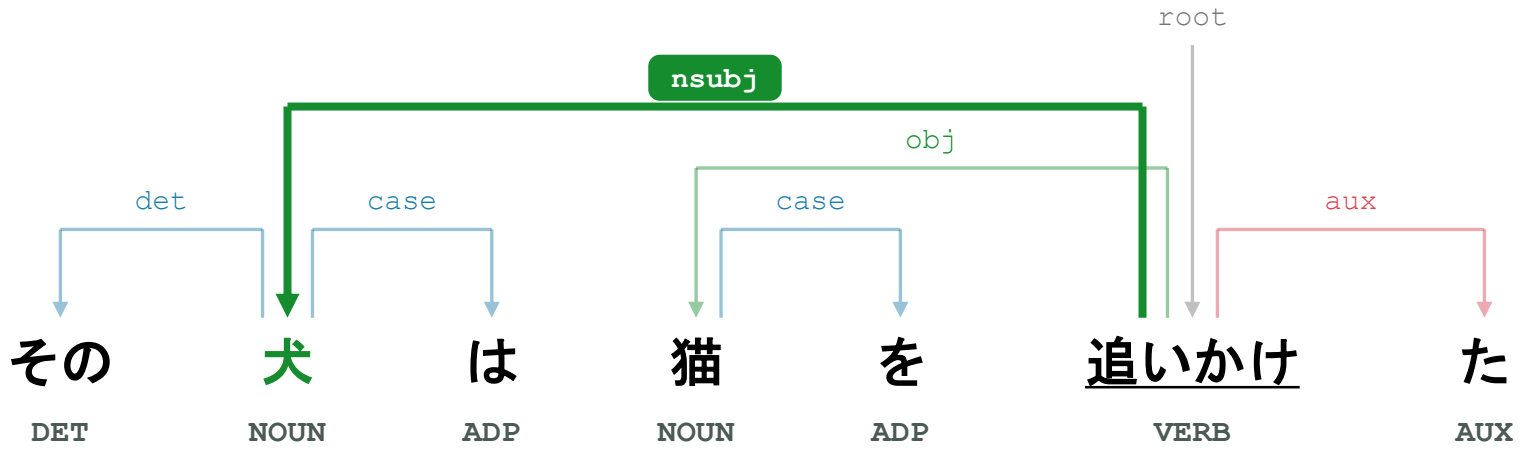
“The dog chased a cat”

CORE ARGUMENTS

`nsubj` `obj` `iobj` `csubj` `ccomp` `xcomp`

nsubj Nominal subject

In Japanese the subject is usually marked by が (or the topic marker は) and precedes the predicate.



‘The dog chased a cat.’

EXAMPLE · DEPENDENT ← HEAD

犬 ← 追いかけ

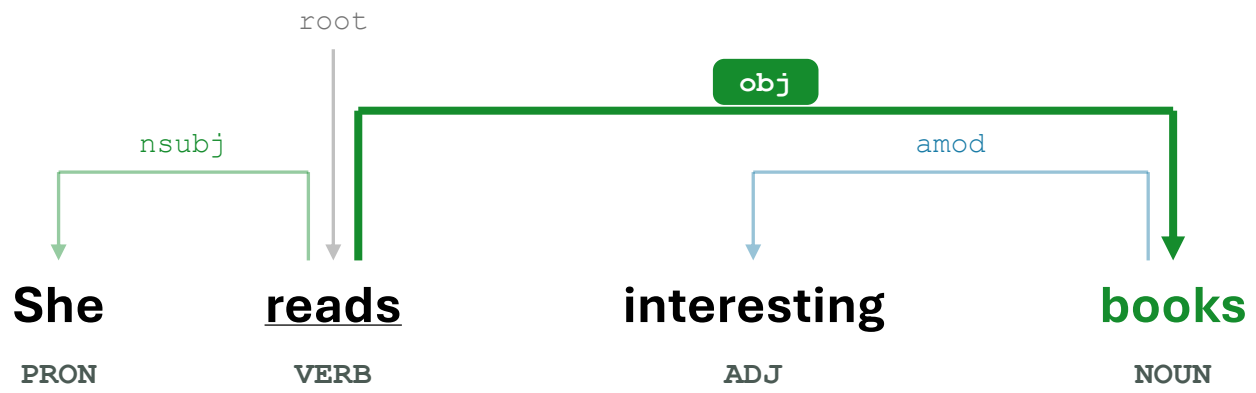
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

obj Object

Direct object: the noun directly affected by the verb's action.



EXAMPLE · DEPENDENT ← HEAD

books ← reads

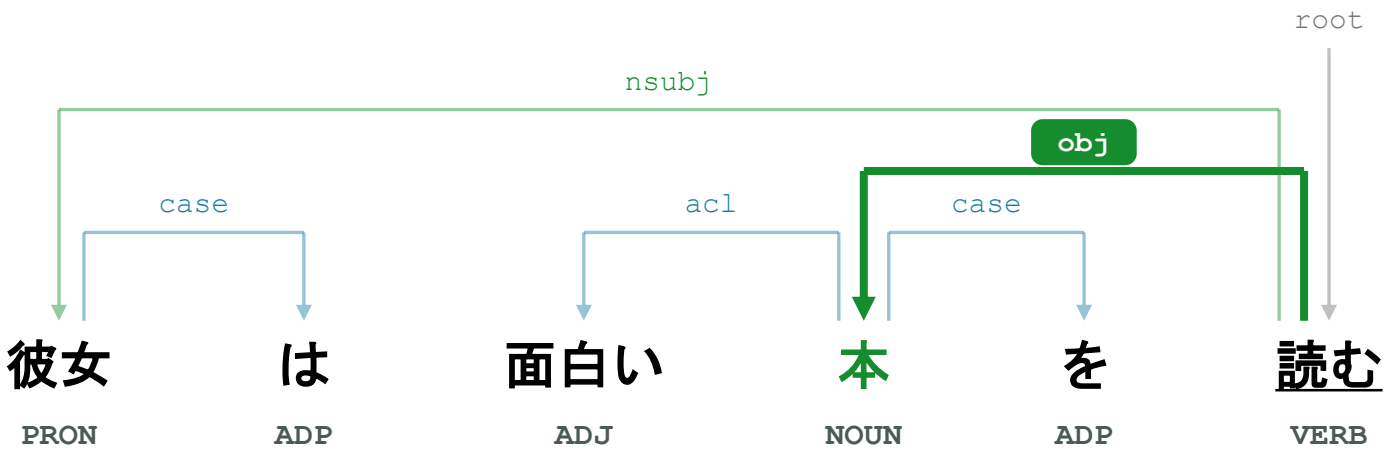
"She reads interesting books"

CORE ARGUMENTS

nsubj | obj | iobj | csubj | ccomp | xcomp

obj Object

The object is marked by を and precedes the verb; the basic word order is subject–object–verb.



‘She reads interesting books.’

EXAMPLE · DEPENDENT ← HEAD

本 ← 読む

Adjectives before nouns are usually analysed as acl in UD Japanese.

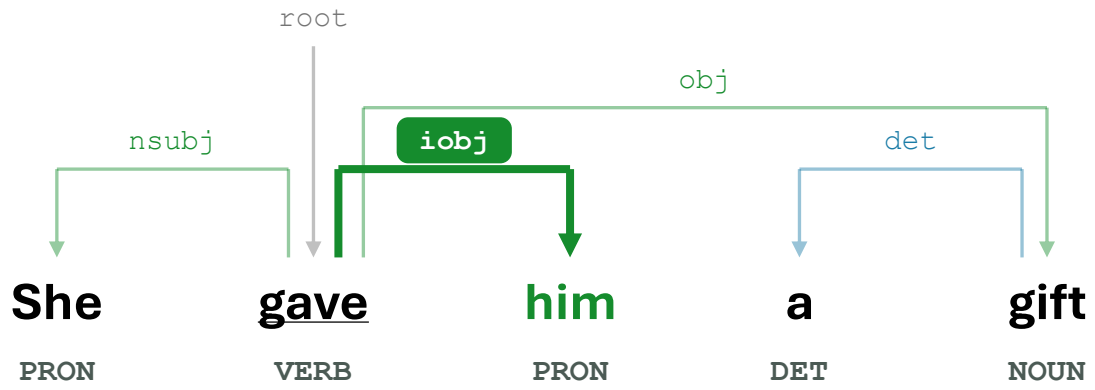
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

iobj Indirect object

Indirect object: recipient or beneficiary, the second object of a verb.



EXAMPLE · DEPENDENT ← HEAD

him ← gave

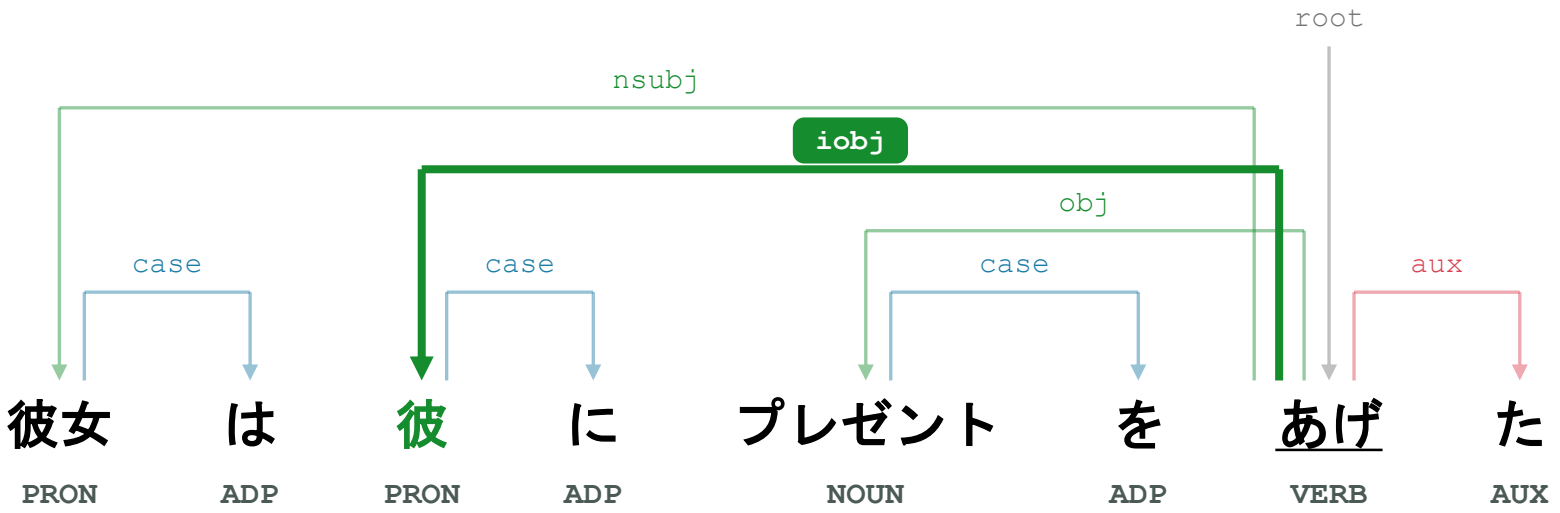
“She gave him a gift”

CORE ARGUMENTS

nsubj obj **iobj** csubj ccomp xcomp

iobj Indirect object

The recipient is usually marked by に; Japanese is head-final, so dependents come before the verb.



'She gave him a gift.'

EXAMPLE · DEPENDENT ← HEAD

彼 ← あげ

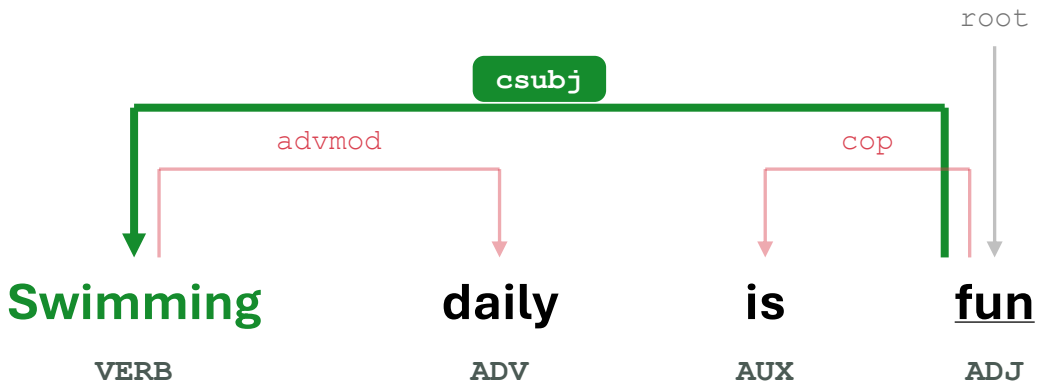
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

csubj Clausal subject

A clause acting as the subject of another clause.



EXAMPLE · DEPENDENT ← HEAD

Swimming ← fun

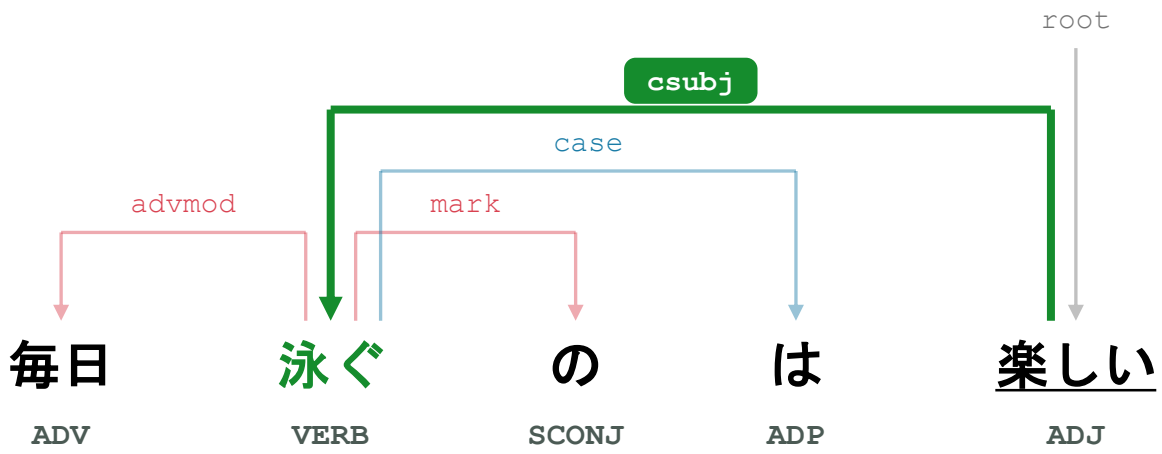
“Swimming daily is fun”

CORE ARGUMENTS

nsubj obj iobj **csubj** ccomp xcomp

csubj Clausal subject

A clause nominalised by の or こと can serve as the subject, followed by は or が.



'Swimming daily is fun.'

EXAMPLE · DEPENDENT ← HEAD

泳ぐ ← 楽しい

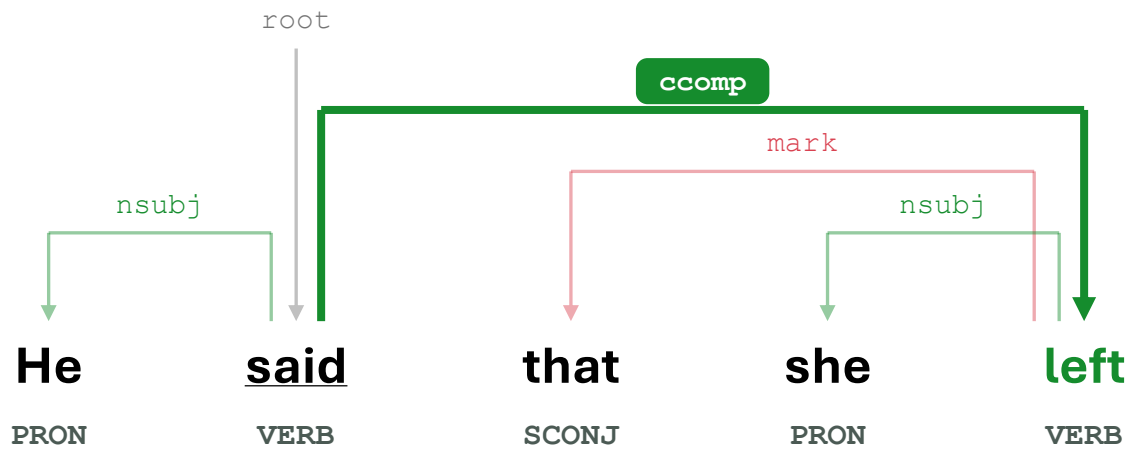
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

ccomp Clausal complement

A clause with its own subject that completes the verb.



EXAMPLE · DEPENDENT ← HEAD

left ← said

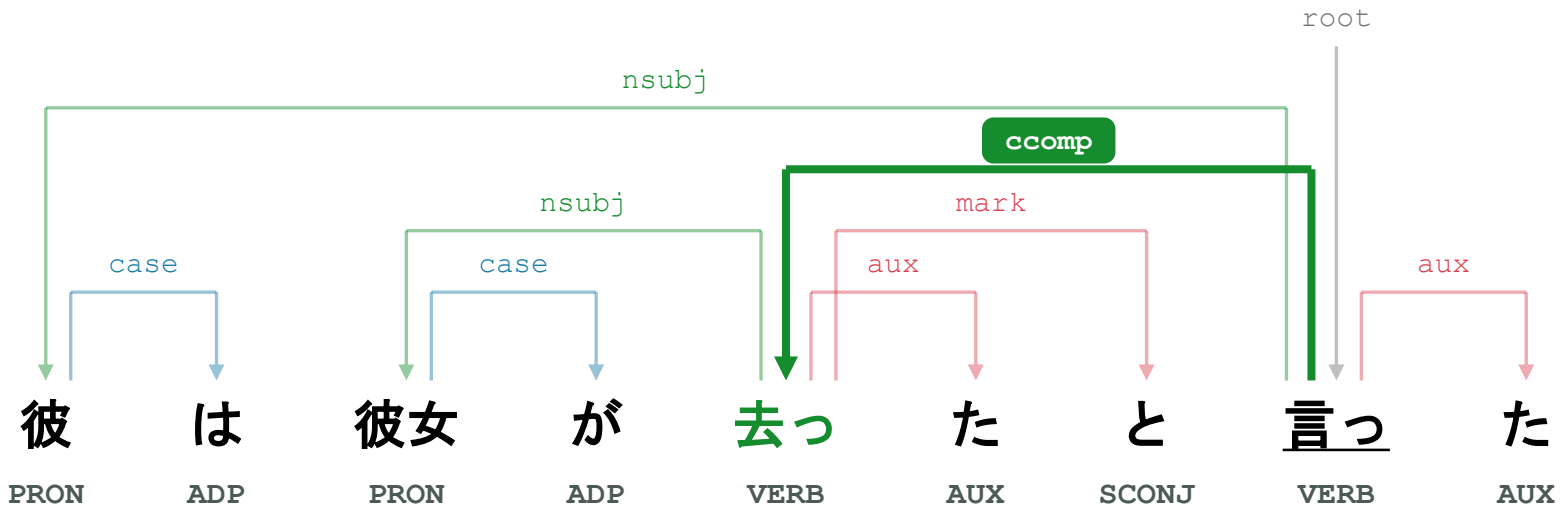
“He said that she left”

CORE ARGUMENTS

nsubj obj iobj csubj ccomp xcomp

ccomp Clausal complement

A reported or quoted clause closed by と precedes the verb of saying or thinking.



'He said that she left.'

EXAMPLE · DEPENDENT ← HEAD

去っ ← 言っ

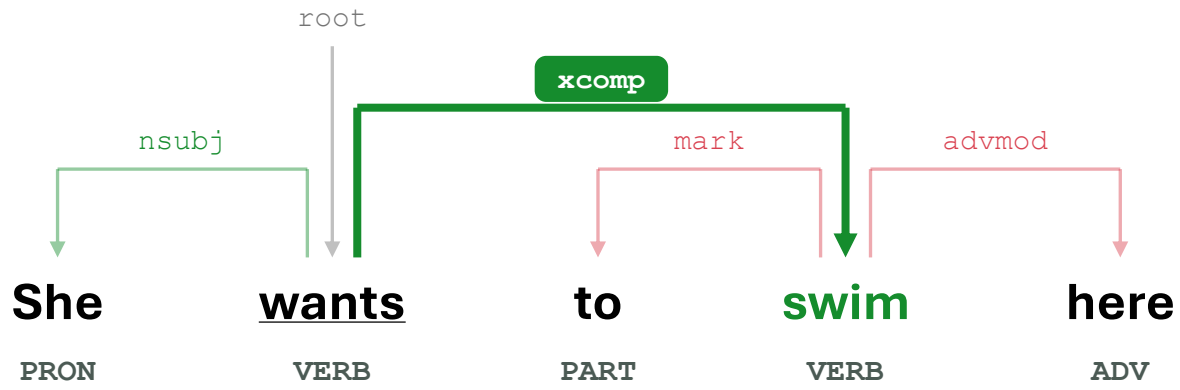
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

xcomp Open clausal complement

A clause that completes the verb and shares the main clause's subject.



EXAMPLE · DEPENDENT ← HEAD

swim ← wants

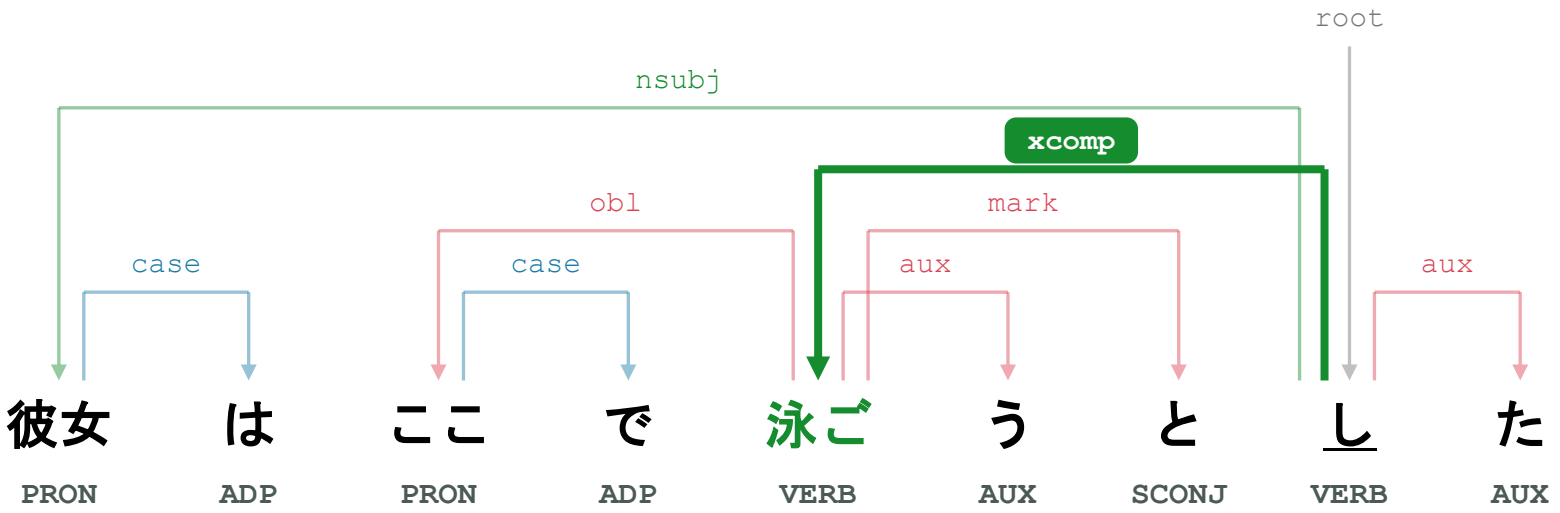
"She wants to swim here"

CORE ARGUMENTS

nsubj obj iobj csubj ccomp **xcomp**

xcomp Open clausal complement

Complements with a controlled subject, such as ～ようとする, attach to the main verb.



‘She tried to swim here.’

EXAMPLE · DEPENDENT ← HEAD

泳ご ← し

‘Wants to swim’ uses the auxiliary ～たい, so ～ようとする (‘tried to’) shows the open complement.

UD JAPANESE

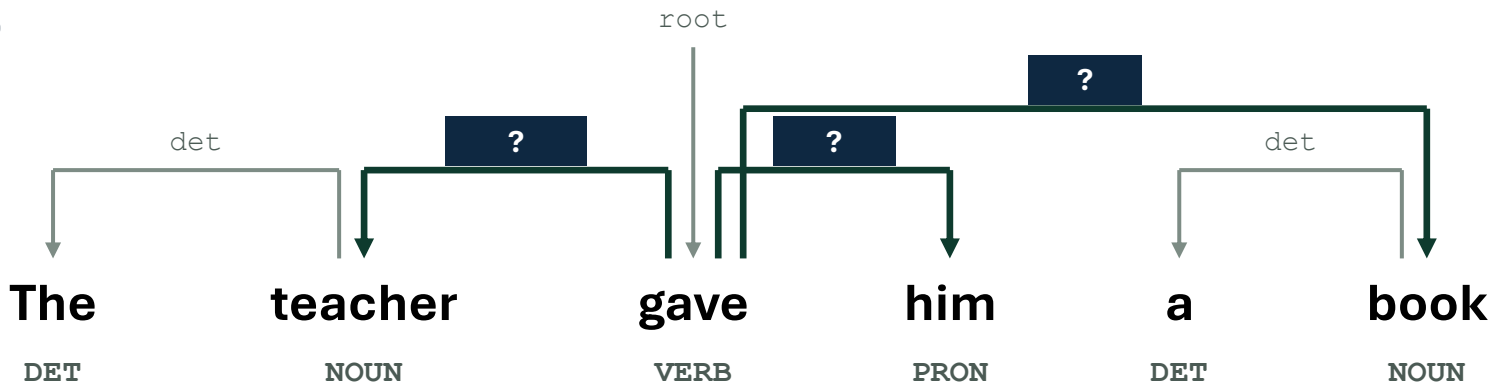
Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

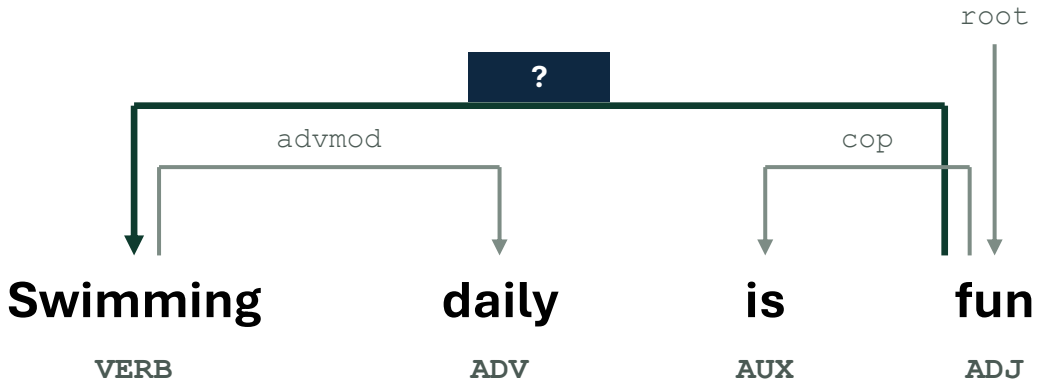
Practice 1 (1/2)

Core Arguments · Click to reveal the hidden relation labels

1



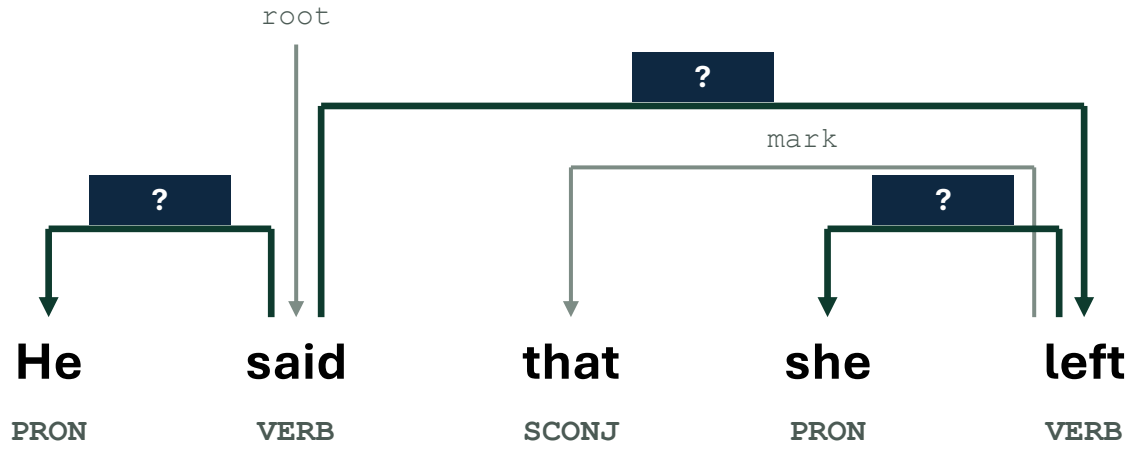
2



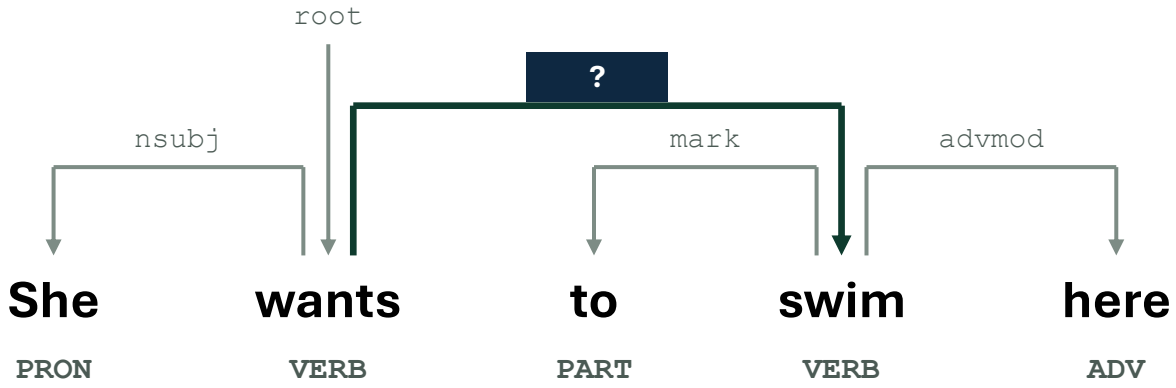
Practice 1 (2/2)

Core Arguments · Click to reveal the hidden relation labels

3

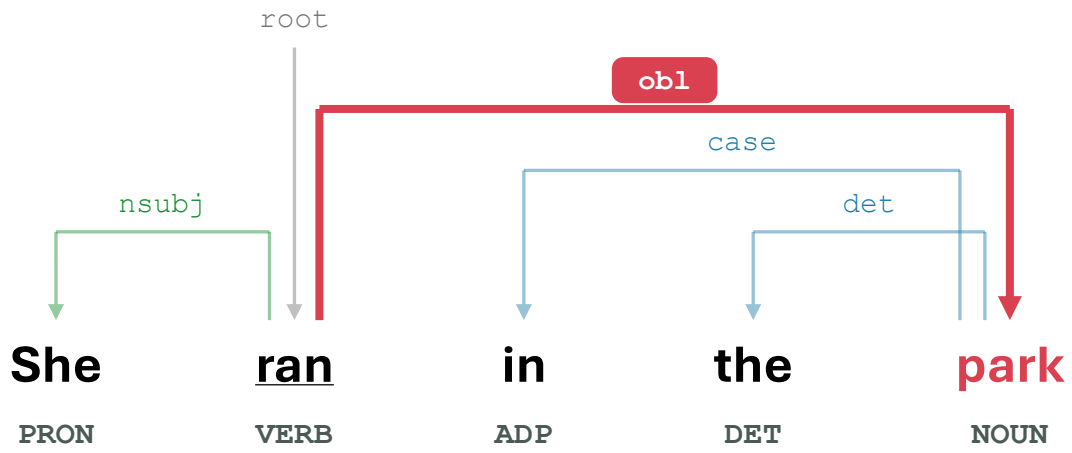


4



obl Oblique nominal

A noun phrase adding place, time, manner or means, not a core argument.



EXAMPLE · DEPENDENT ← HEAD

park ← ran

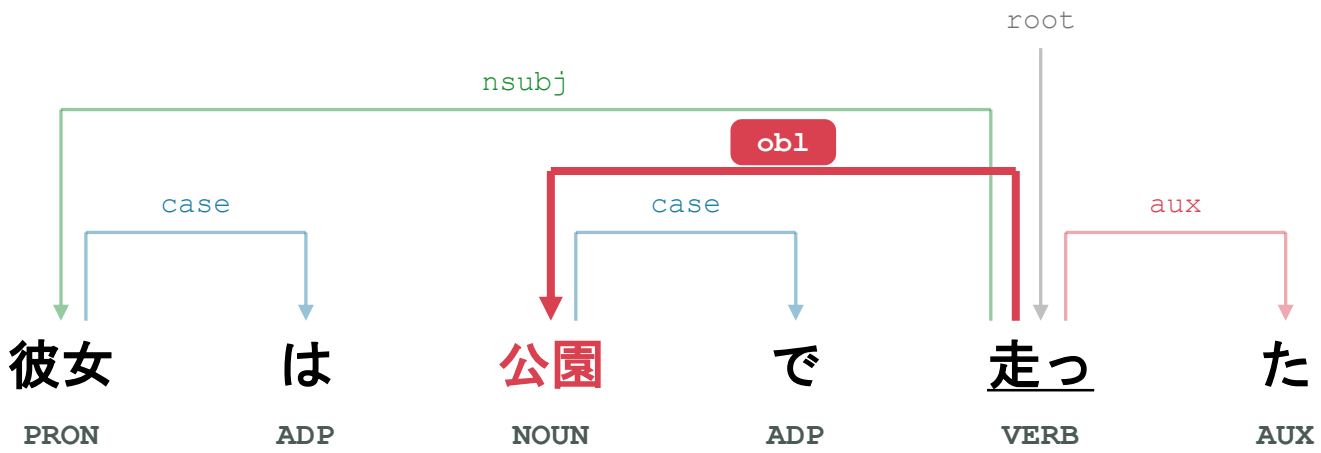
“She ran in the park”

NON-CORE DEPENDENTS

obl vocative expl dislocated advcl advmod discourse aux cop mark

obl Oblique nominal

Place, time and instrument phrases take particles such as で, に or から and precede the verb.



‘She ran in the park.’

EXAMPLE · DEPENDENT ← HEAD

公園 ← 走っ

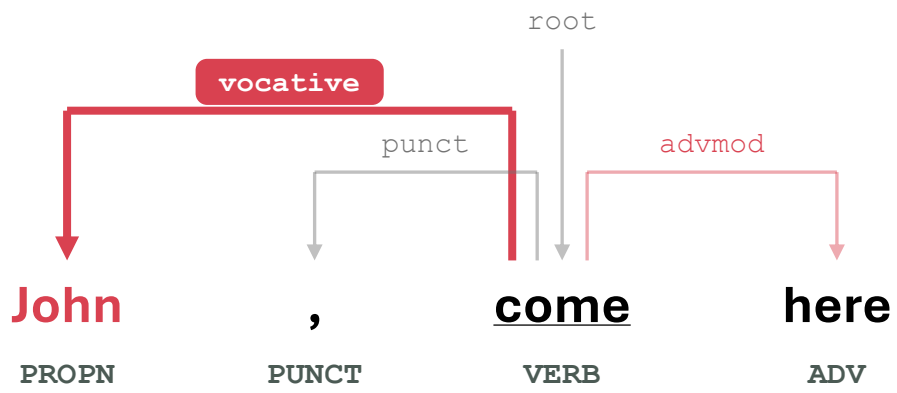
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

vocative Vocative

A noun used to address someone directly.



EXAMPLE · DEPENDENT ← HEAD

John ← come

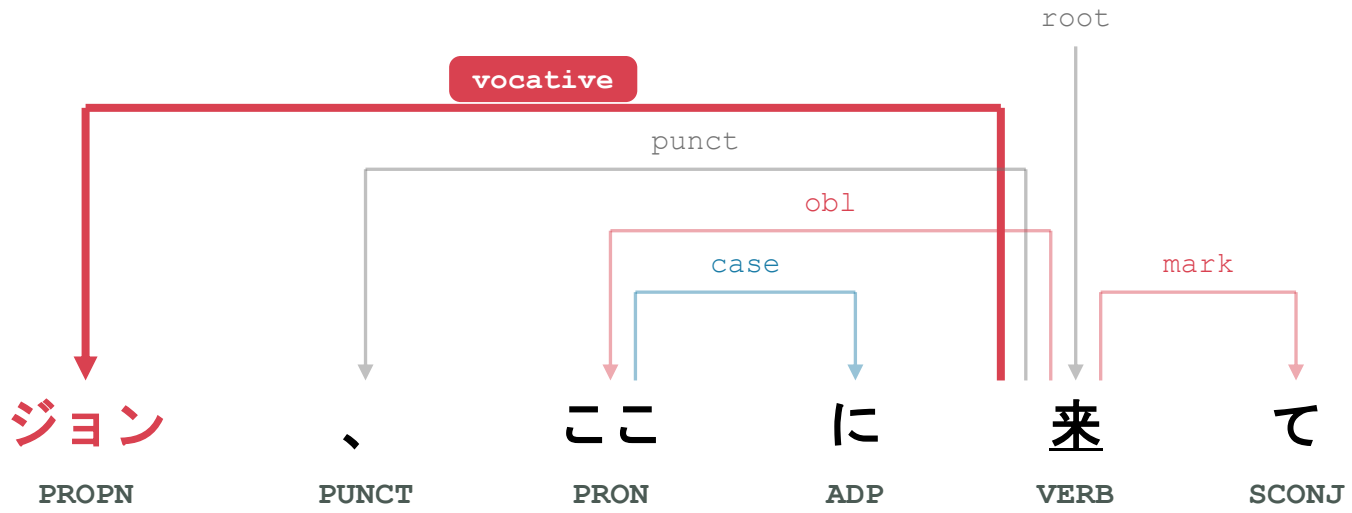
“John, come here”

NON-CORE DEPENDENTS

obl | vocative | expl | dislocated | advcl | advmod | discourse | aux | cop | mark

vocative Vocative

A name or title used to address someone is set off by a comma and attaches to the predicate.



‘John, come here.’

EXAMPLE · DEPENDENT ← HEAD

ジョン ← 来

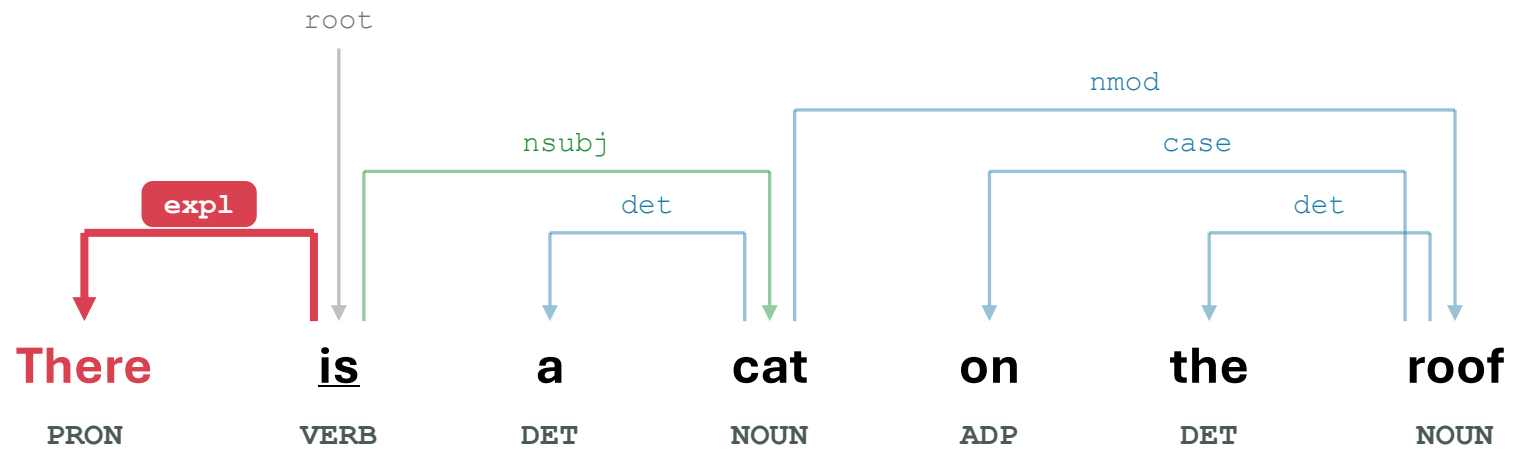
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

expl Expletive

A dummy pronoun filling the subject slot, like ‘there’ or ‘it’.



EXAMPLE · DEPENDENT ← HEAD

There ← is

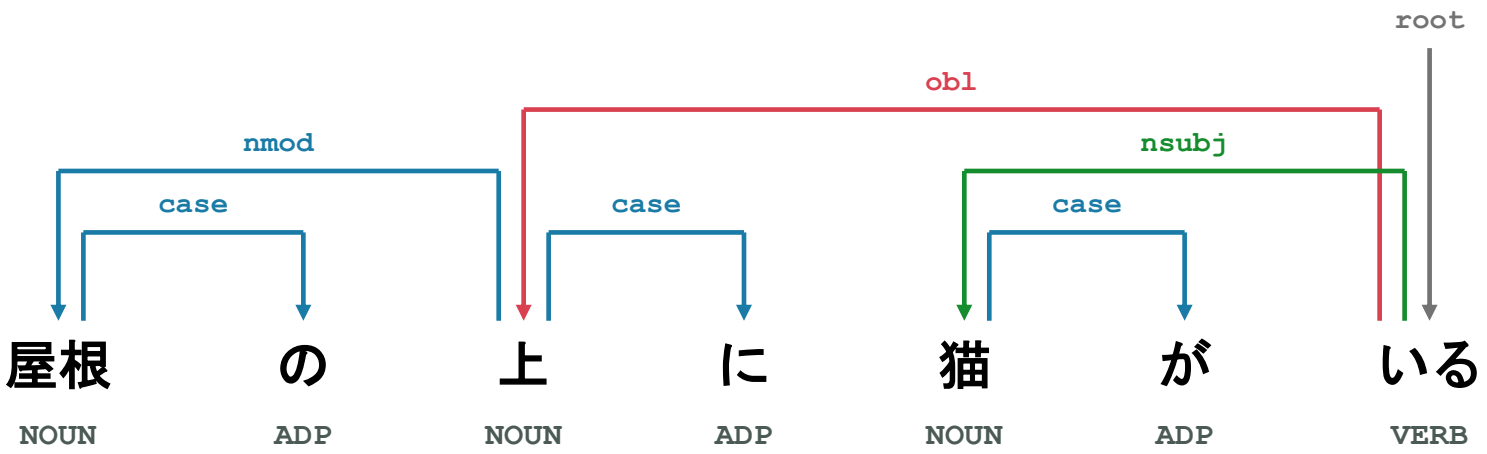
“There is a cat on the roof”

NON-CORE DEPENDENTS

obl · vocative · expl · dislocated · advcl · advmod · discourse · aux · cop · mark

Column Expletives across languages

English fills the subject slot with ‘there’; Japanese uses an ordinary nsubj, so no expl is needed.



‘There is a cat on the roof.’

ENGLISH vs JAPANESE

English: There ← is (expl)

Japanese: 猫 ← いる (nsubj), no expl

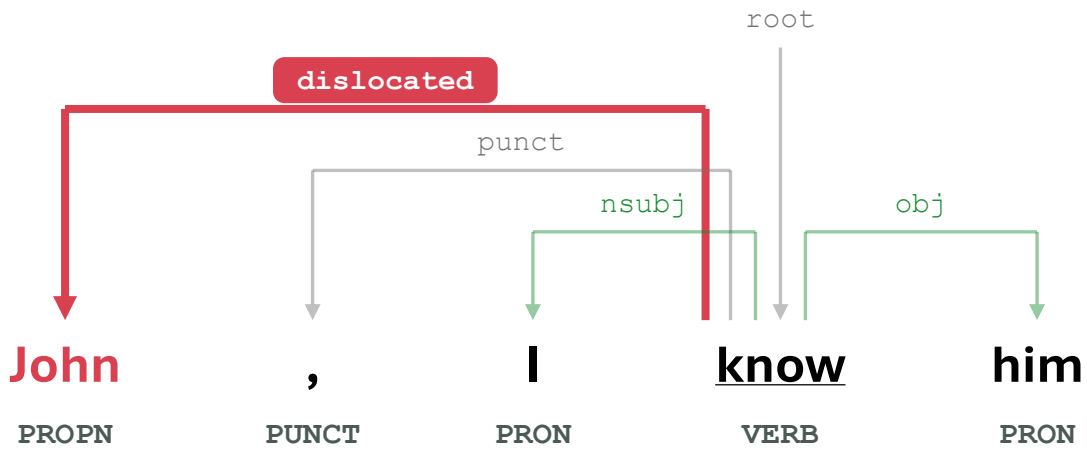
UD JAPANESE

Not used in Japanese: the subject slot needs no dummy word.

Hand-annotated after UD Japanese conventions.

dislocated Dislocated elements

A fronted or trailing topic outside the clause, echoed by a pronoun.



EXAMPLE · DEPENDENT ← HEAD

John ← know

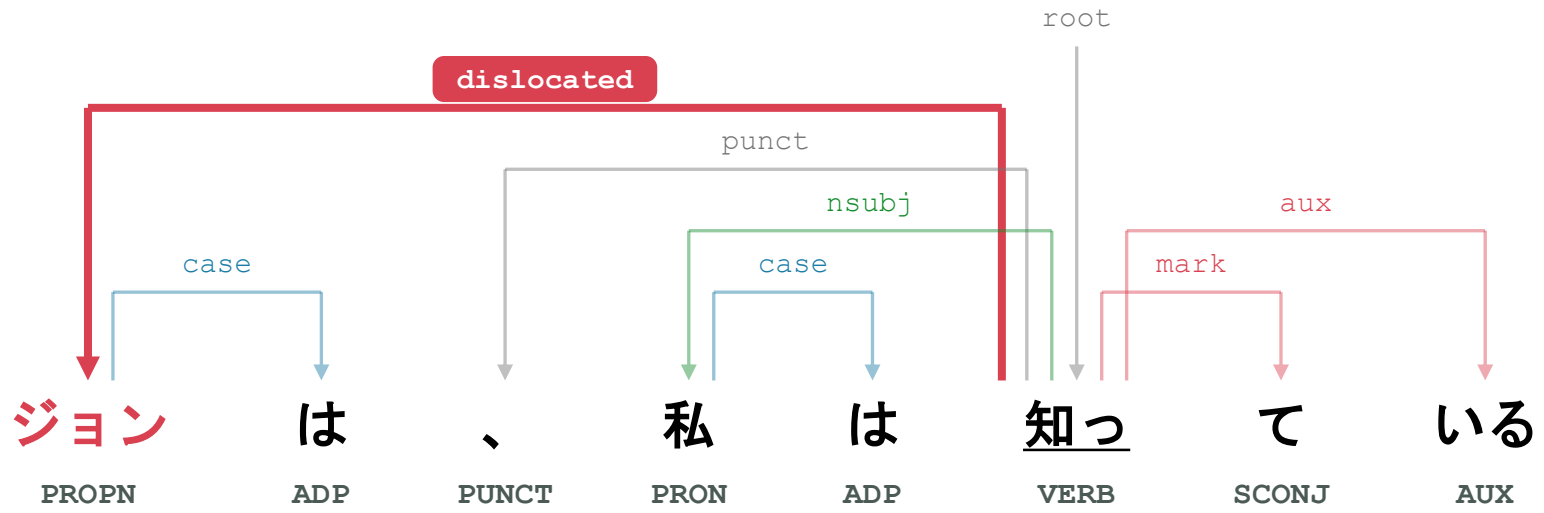
“John, I know him”

NON-CORE DEPENDENTS

obl vocative expl dislocated advcl advmod discourse aux cop mark

dislocated Dislocated elements

A topic phrase with は and a comma stands outside the clause and is resumed inside it.



'John, I know (him).'

EXAMPLE · DEPENDENT ← HEAD

ジョン ← 知っ

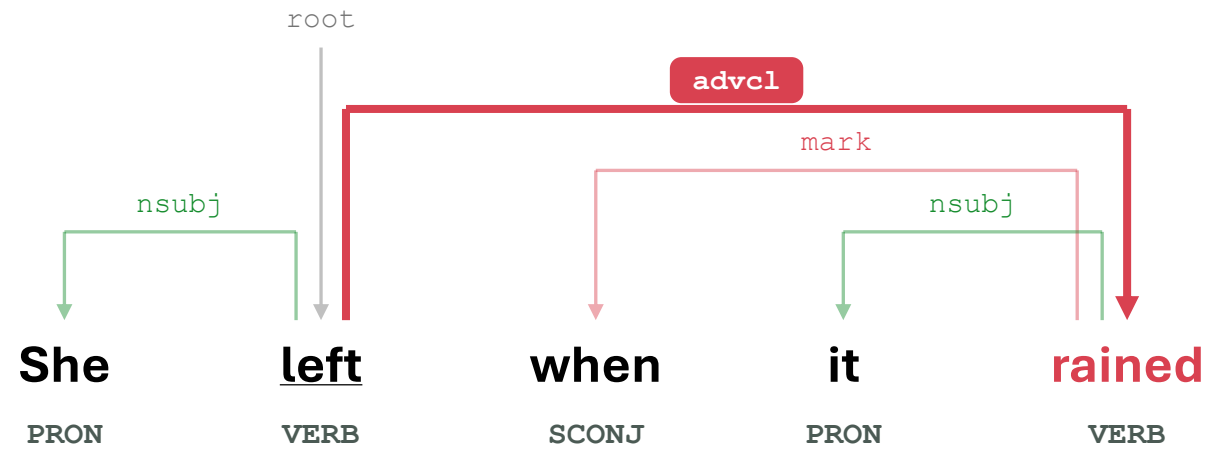
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

advcl Adverbial clause modifier

A subordinate clause modifying a verb: time, reason, condition.



EXAMPLE · DEPENDENT ← HEAD

rained ← left

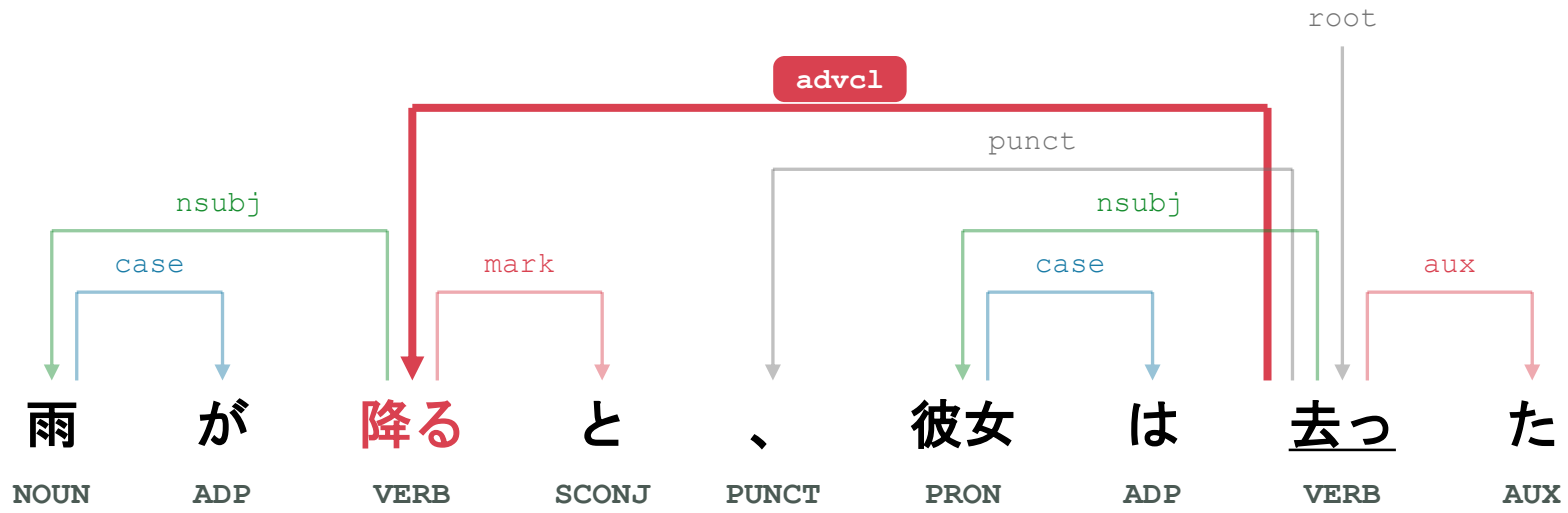
“She left when it rained”

NON-CORE DEPENDENTS

obl vocative expl
dislocated advcl advmod
discourse aux cop mark

advcl Adverbial clause modifier

Subordinate clauses end in conjunctive particles like と, から, ので or ば and precede the main clause.



‘When it rained, she left.’

EXAMPLE · DEPENDENT ← HEAD

降る ← 去っ

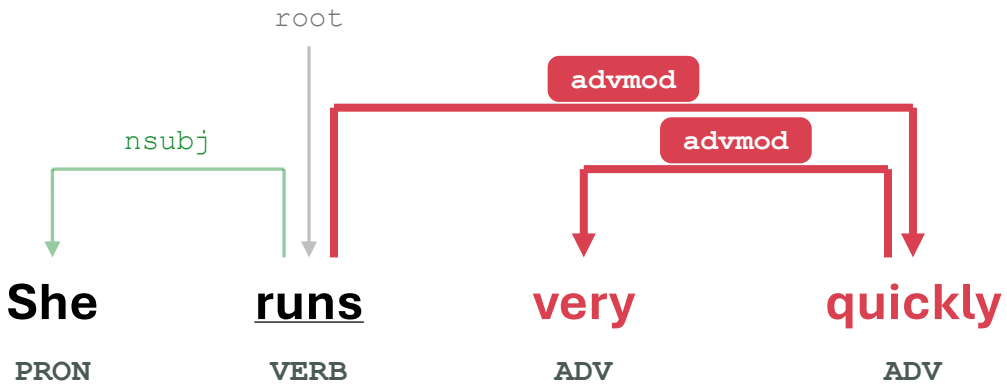
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

advmod Adverbial modifier

An adverb (or adverbial word) modifying a verb, adjective or adverb.



EXAMPLE · DEPENDENT ← HEAD

very ← quickly **quickly ← runs**

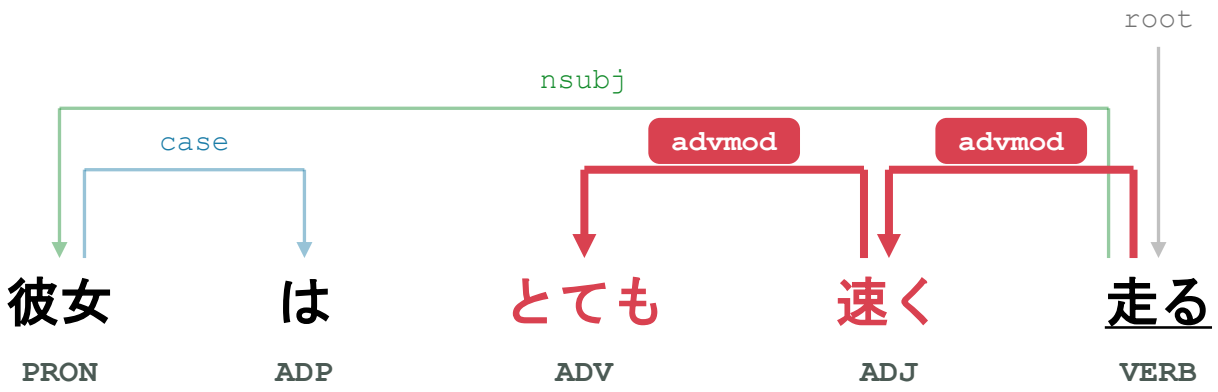
“She runs very quickly”

NON-CORE DEPENDENTS

obl vocative expl
dislocated advcl advmod
discourse aux cop mark

advmod Adverbial modifier

Adverbs such as とても or 速く modify the verb or adjective that follows.



‘She runs very quickly.’

EXAMPLE · DEPENDENT ← HEAD

とても ← 速く　速く ← 走る

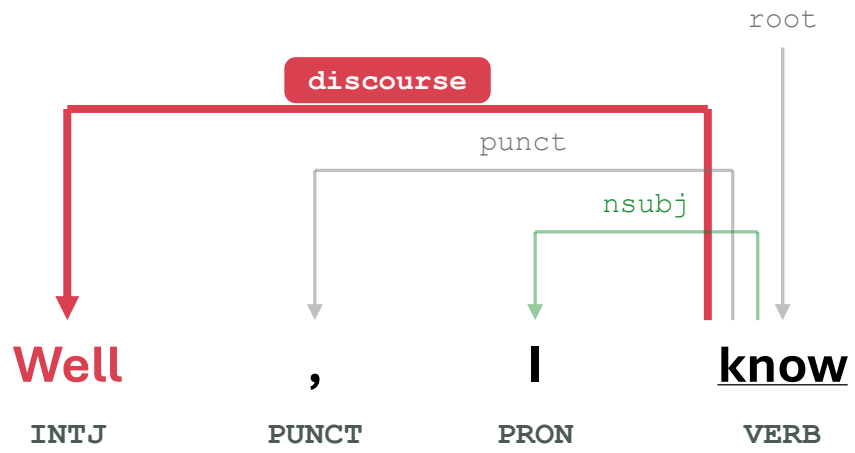
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

discourse Discourse element

Interjections and fillers that add tone, not meaning: well, oh, lol.



EXAMPLE · DEPENDENT ← HEAD

Well ← know

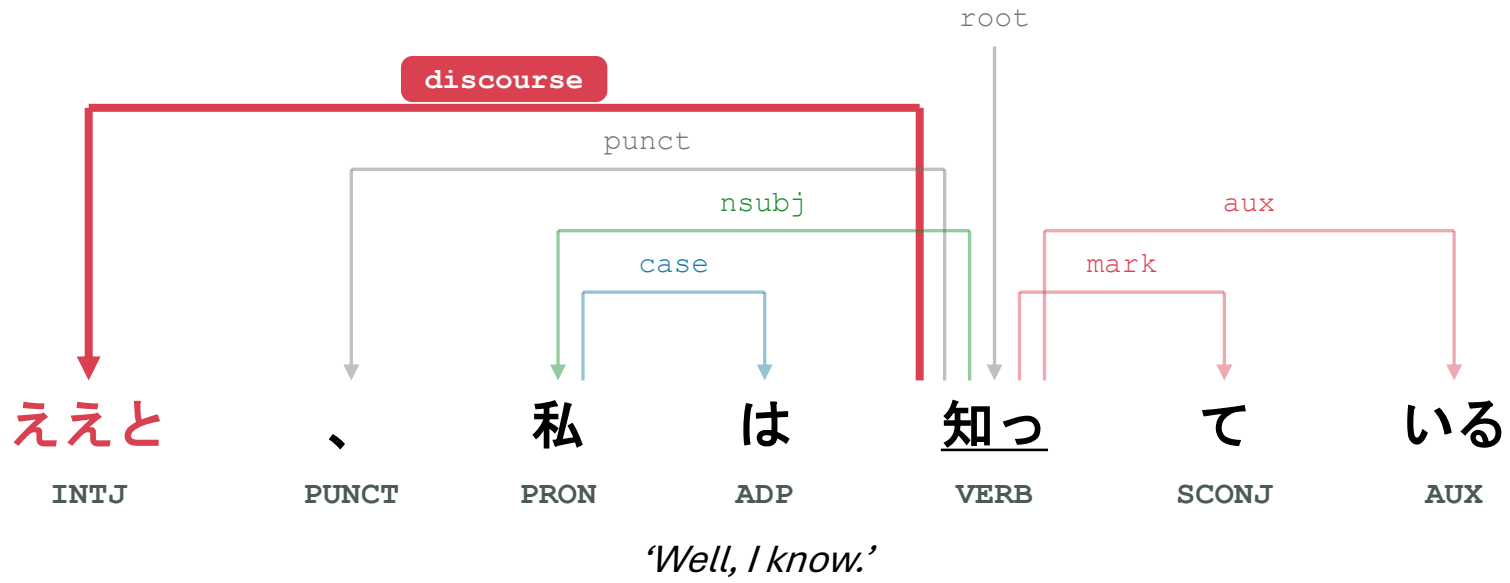
“Well, I know”

NON-CORE DEPENDENTS

obl vocative expl dislocated advcl advmod discourse aux cop mark

discourse Discourse element

Fillers and interjections such as ええと and えっと open the utterance.



‘Well, I know.’

EXAMPLE · DEPENDENT ← HEAD

ええと ← 知っ

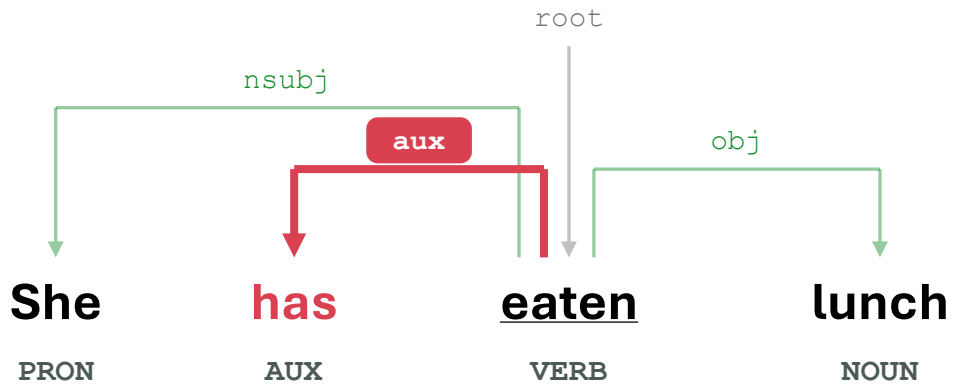
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

aux Auxiliary

A helper verb marking tense, aspect, mood or voice.



EXAMPLE · DEPENDENT ← HEAD

has ← eaten

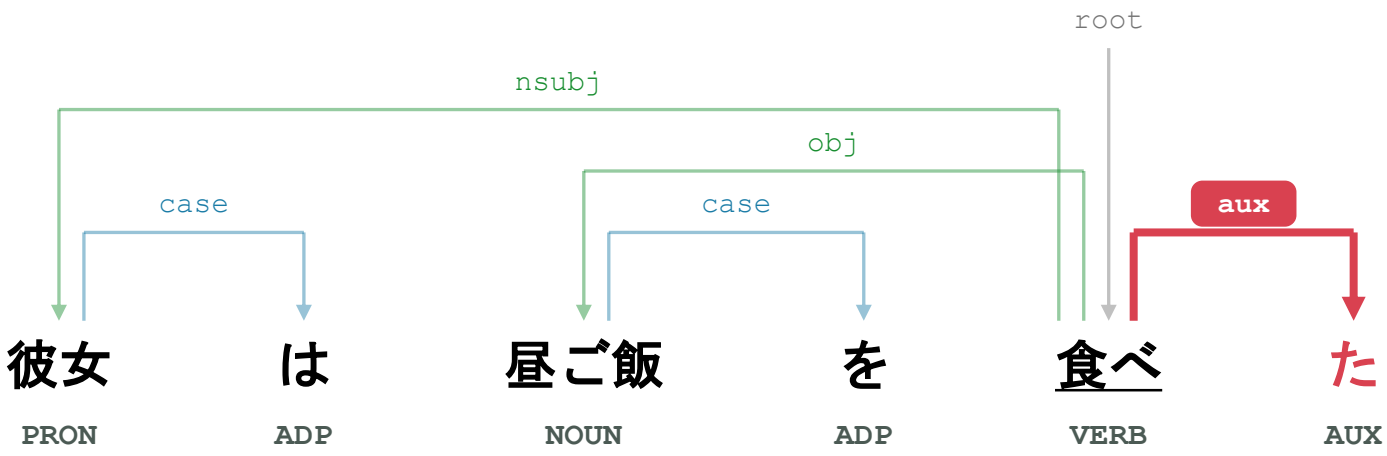
“She has eaten lunch”

NON-CORE DEPENDENTS

obl vocative expl dislocated advcl advmod discourse **aux** cop mark

aux Auxiliary

Tense, negation and politeness are separate auxiliaries after the verb stem: た, ない, ます.



‘She has eaten lunch.’

EXAMPLE · DEPENDENT ← HEAD

た ← 食べ

The past auxiliary た also covers the English present perfect.

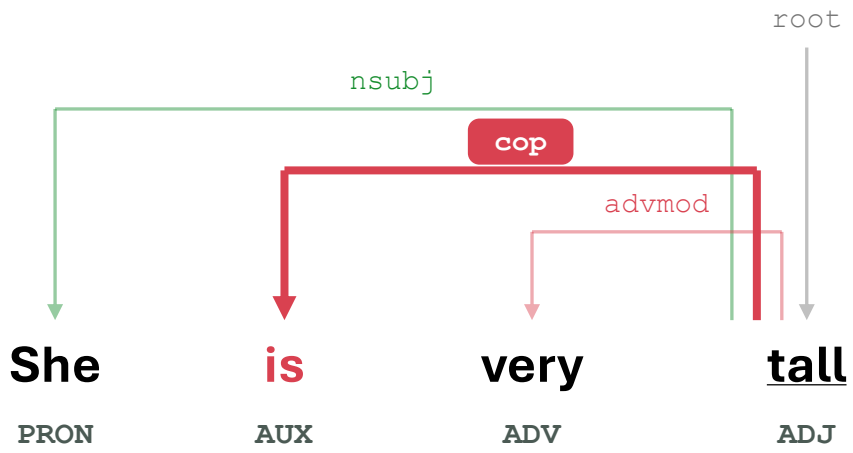
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

cop Copula

The verb ‘be’ linking a subject to a non-verbal predicate.



EXAMPLE · DEPENDENT ← HEAD

is ← tall

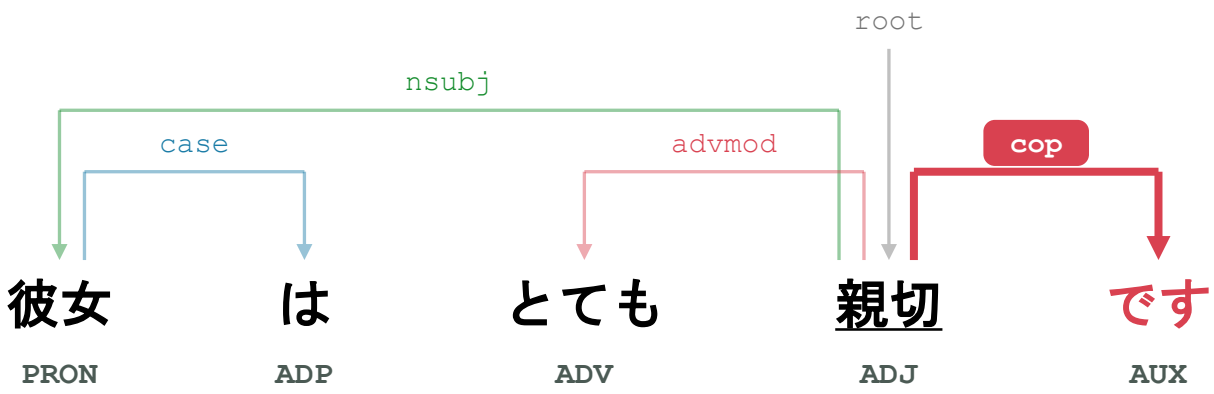
“She is very tall”

NON-CORE DEPENDENTS

obl vocative expl dislocated advcl advmod discourse aux **cop** mark

cop Copula

The copula だ / です follows a noun or na-adjective; i-adjectives like 高い need no copula.



‘She is very kind.’

EXAMPLE · DEPENDENT ← HEAD

です ← 親切

‘Tall’ (背が高い) is an i-adjective and takes no copula, so a na-adjective is used.

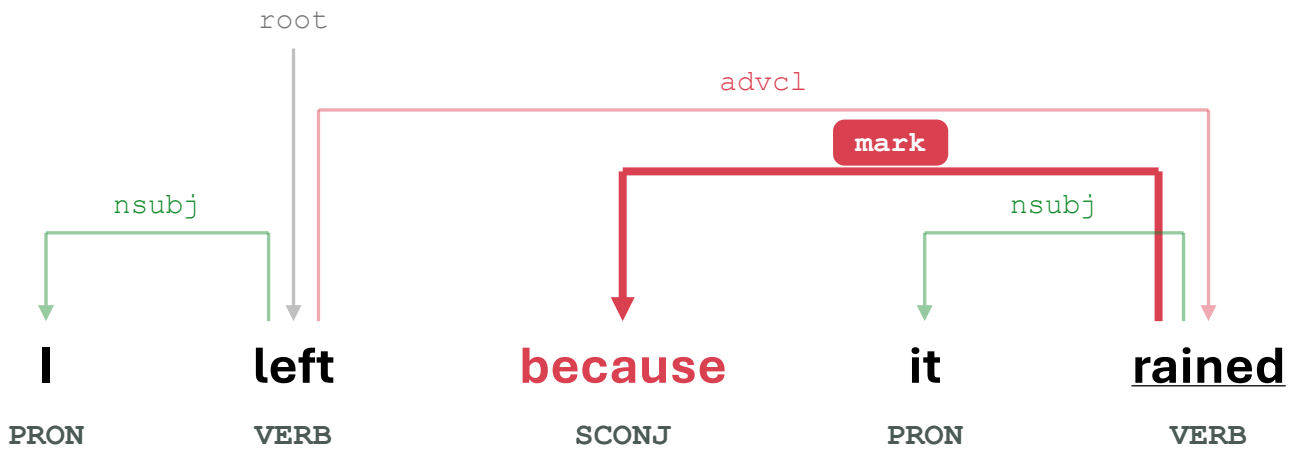
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

`mark` Marker

A word introducing a subordinate clause: that, because, if, to.



EXAMPLE · DEPENDENT ← HEAD

because ← rained

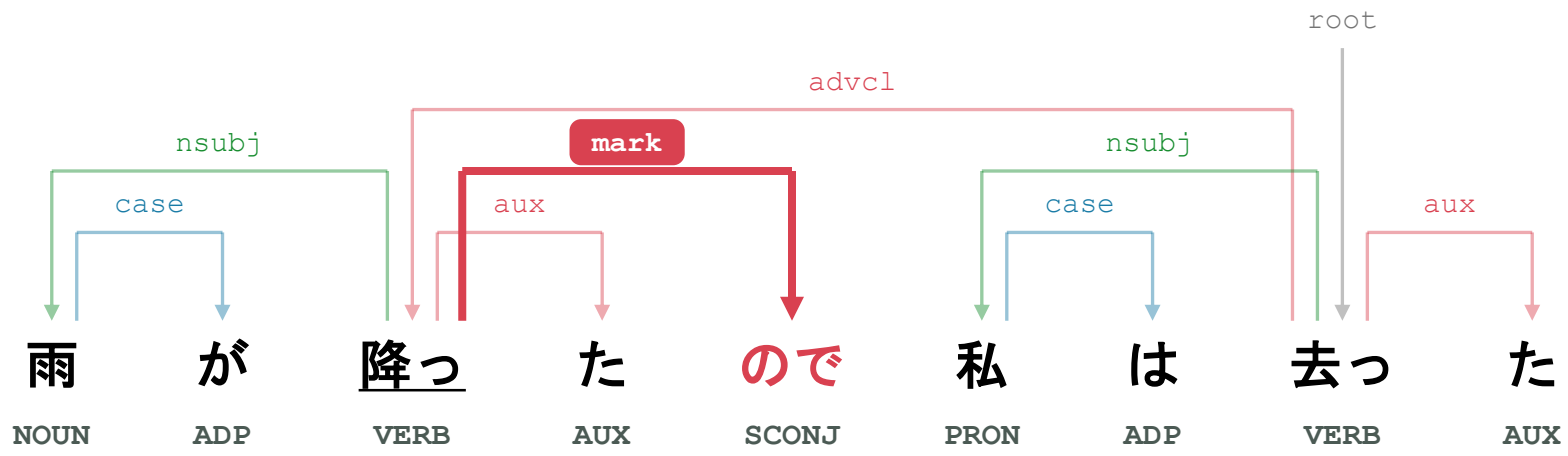
“I left because it rained”

NON-CORE DEPENDENTS

`obl` `vocative` `expl` `dislocated` `advcl` `advmod` `discourse` `aux` `cop` `mark`

mark Marker

Subordinating particles such as ので, から and て follow the clause they introduce.



‘I left because it rained.’

EXAMPLE · DEPENDENT ← HEAD

ので ← 降っ

UD JAPANESE

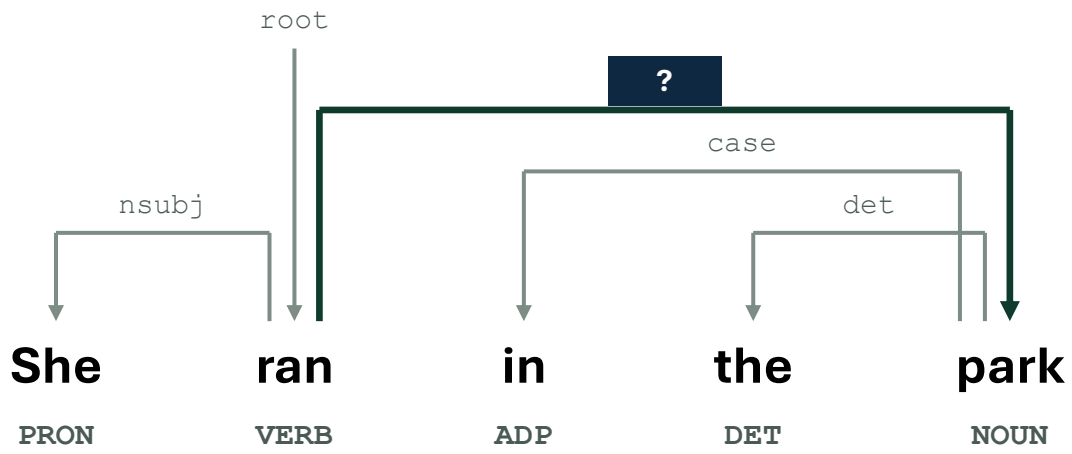
Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

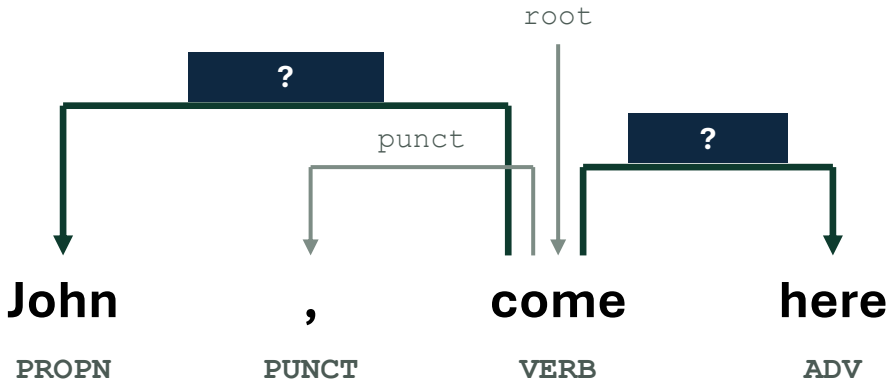
Practice 2 (1/3)

Non-core Dependents · Click to reveal the hidden relation labels

1



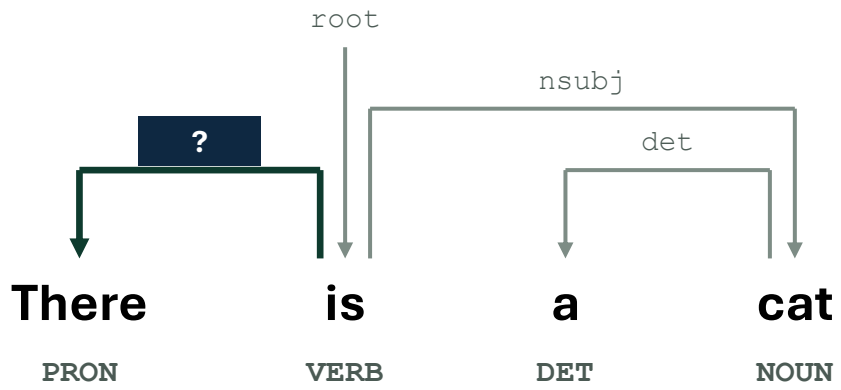
2



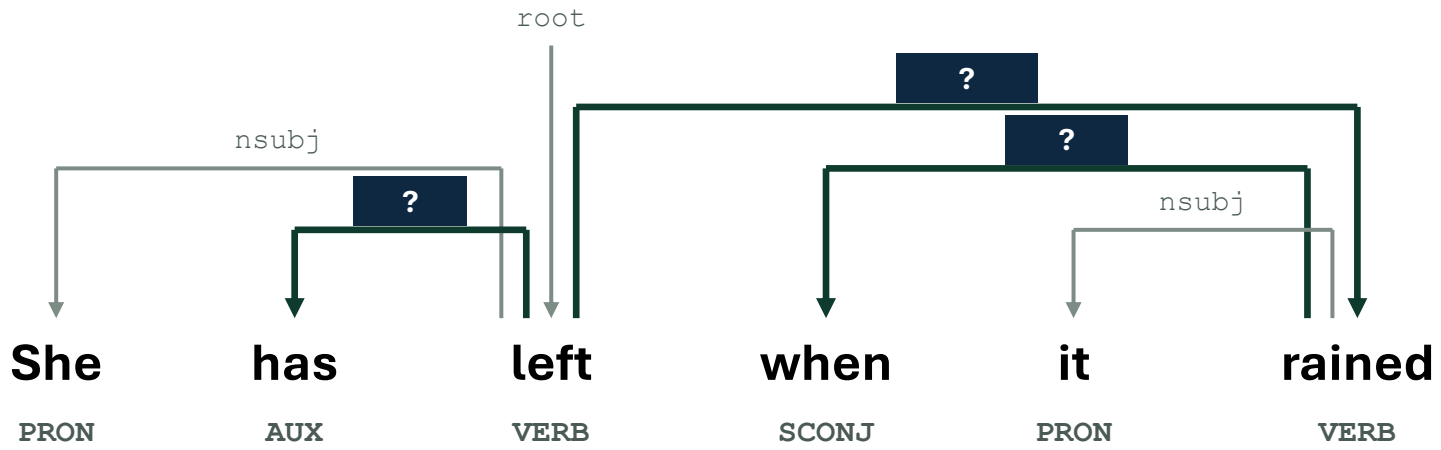
Practice 2 (2/3)

Non-core Dependents · Click to reveal the hidden relation labels

3



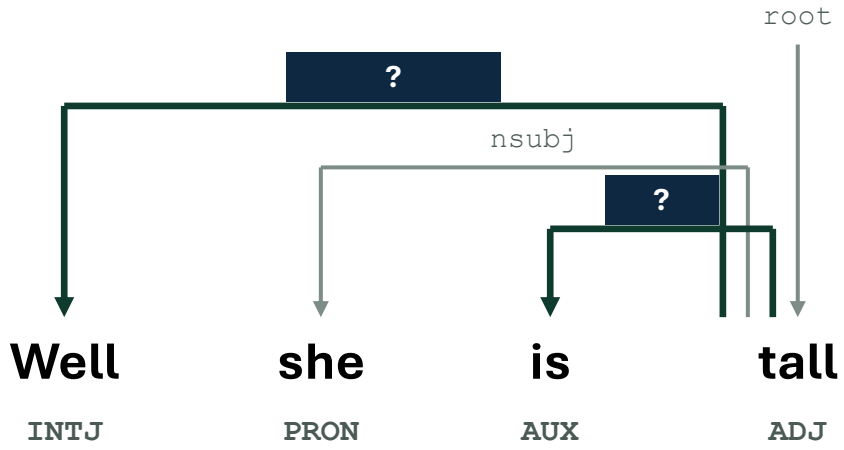
4



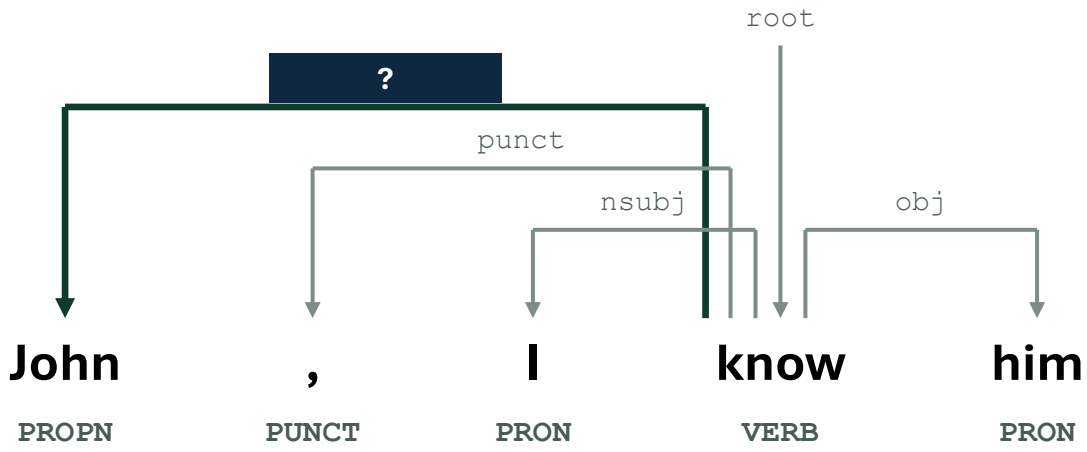
Practice 2 (3/3)

Non-core Dependents · Click to reveal the hidden relation labels

5

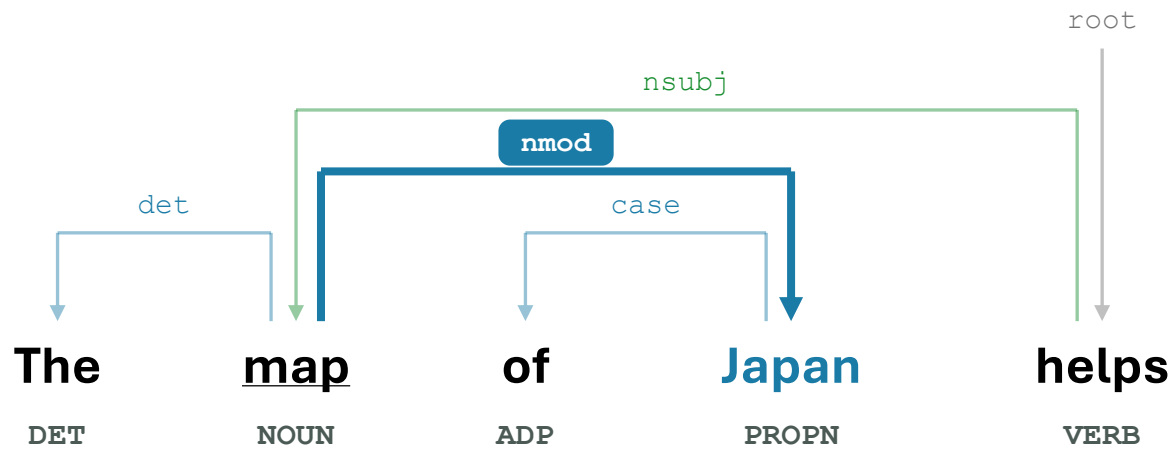


6



nmod Nominal modifier

A noun that modifies another noun, often via a preposition.



EXAMPLE · DEPENDENT ← HEAD

Japan ← map

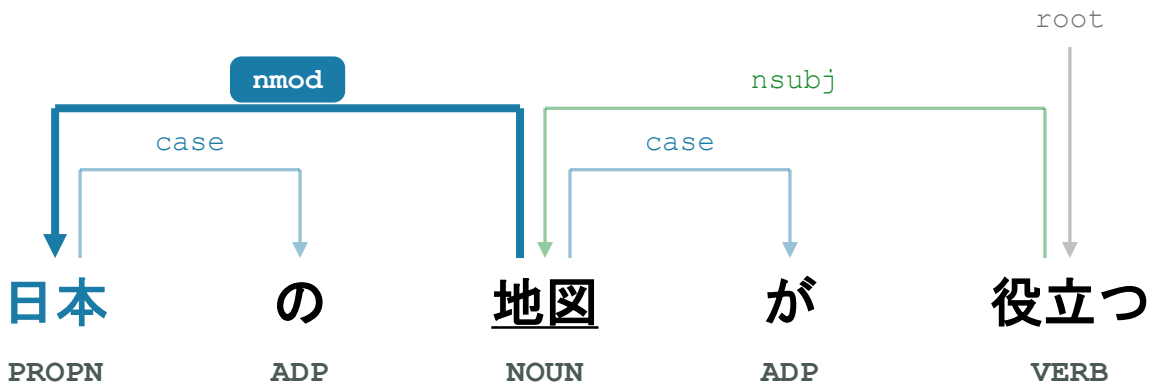
“The map of Japan helps”

NOMINAL DEPENDENTS

nmod appos nummod acl amod det clf case

nmod Nominal modifier

The genitive の attaches to the possessor noun, which precedes the head noun.



‘The map of Japan helps.’

EXAMPLE · DEPENDENT ← HEAD

日本 ← 地図

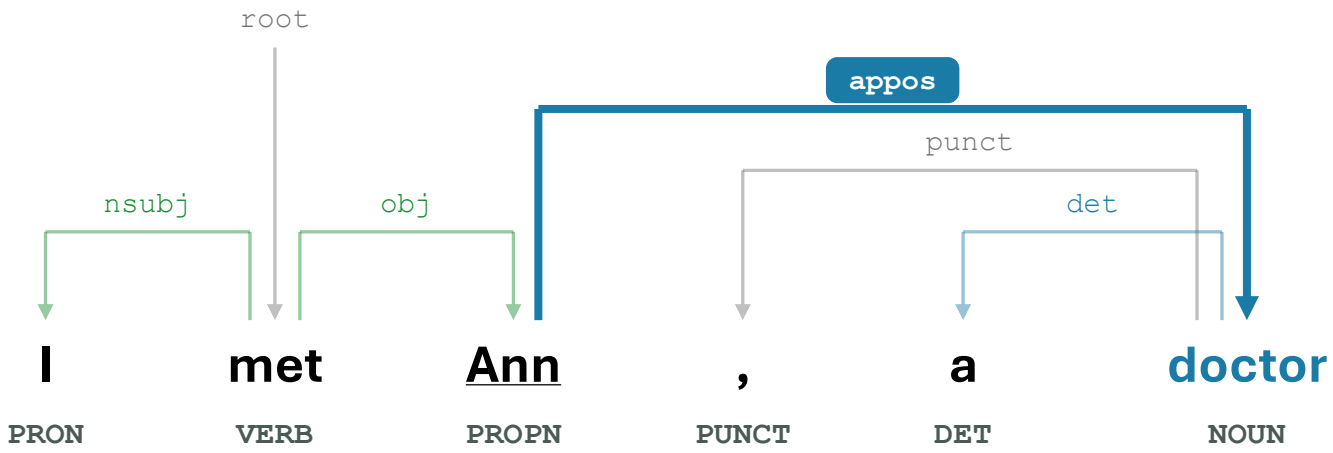
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

appos Appositional modifier

A noun phrase that renames or explains the noun beside it.



EXAMPLE · DEPENDENT ← HEAD

doctor ← Ann

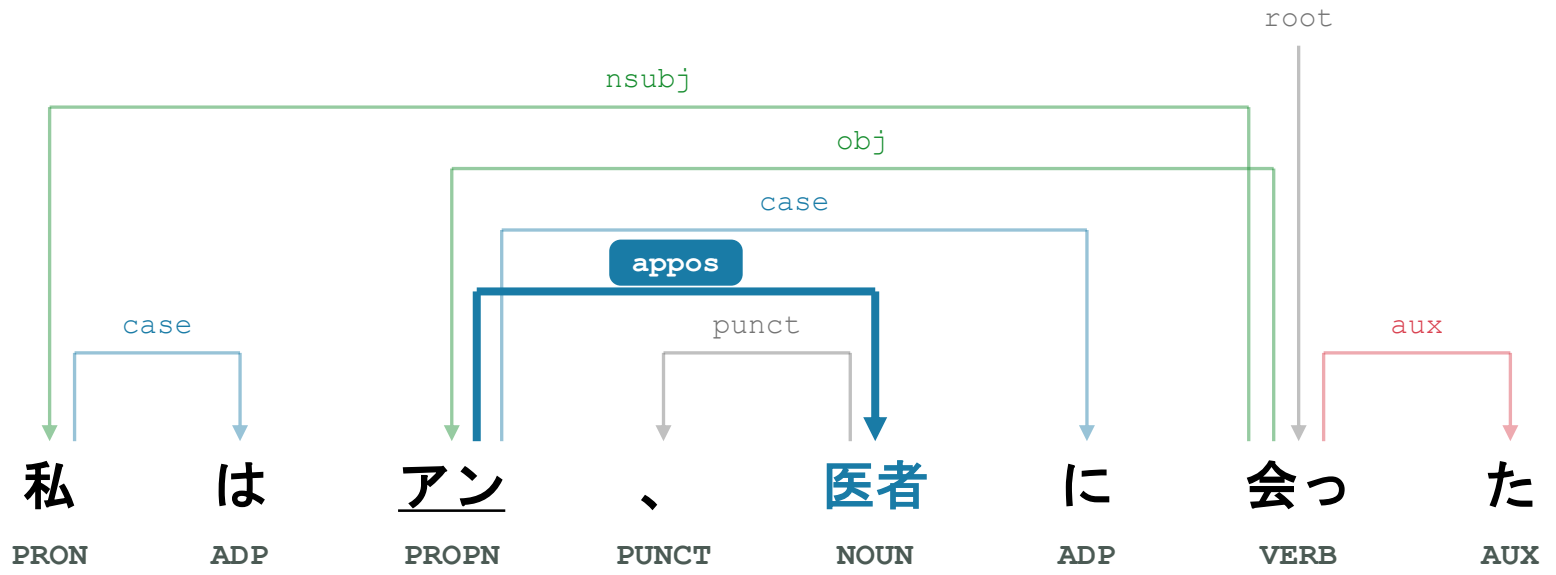
“I met Ann, a doctor”

NOMINAL DEPENDENTS

nmod | **appos** | nummod | acl | amod | det | clf | case

appos Appositional modifier

A noun phrase can be placed beside another noun to rename it, set off by a comma.



‘I met Ann, a doctor.’

EXAMPLE · DEPENDENT ← HEAD

医者 ← アン

Japanese usually writes ‘医者のアン’, analysed as nmod; this comma form is constructed.

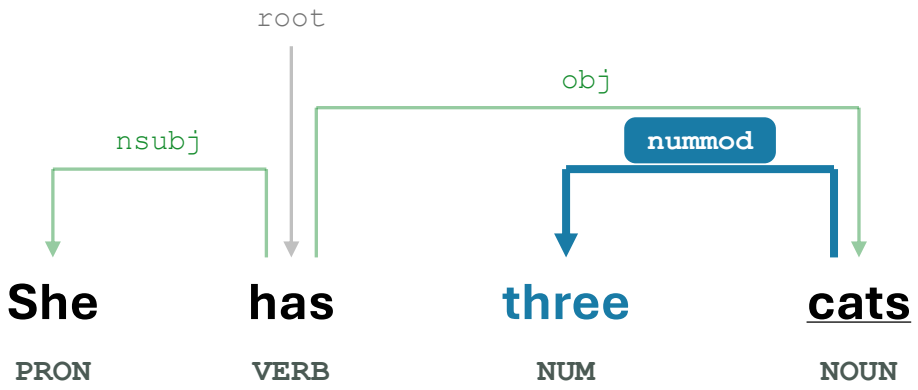
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

nummod Numeric modifier

A number that modifies a noun.



EXAMPLE · DEPENDENT ← HEAD

three ← cats

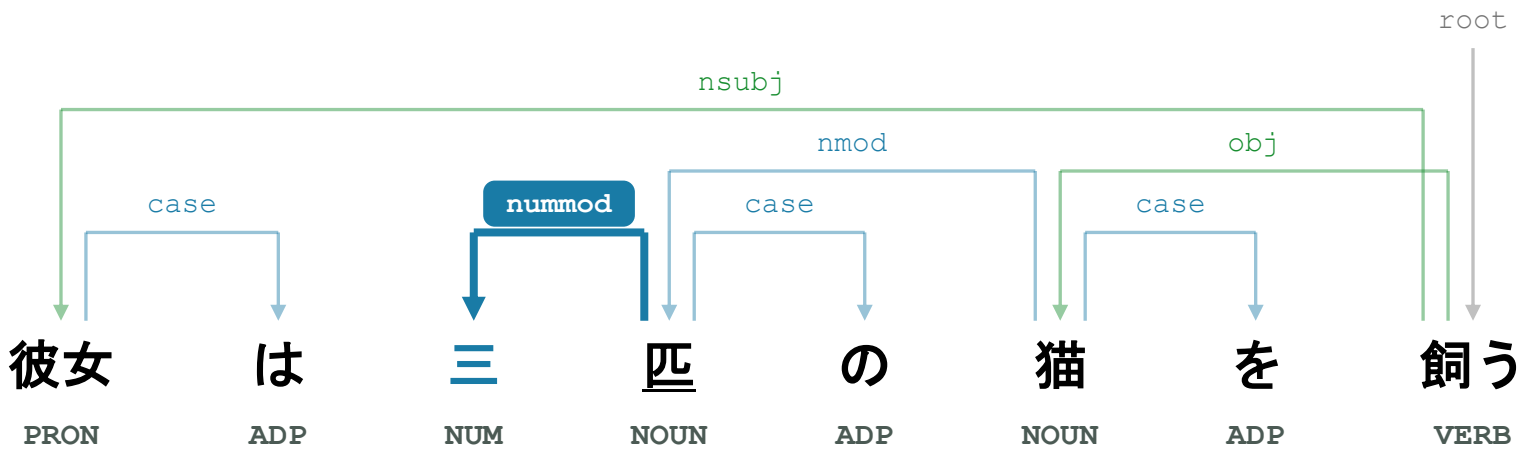
“She has three cats”

NOMINAL DEPENDENTS

nmod · appos · nummod · acl · amod · det · clf · case

nummod Numeric modifier

Numerals combine with counters (つ, 匹, 人) to form a measure phrase before the noun.



‘She keeps three cats.’

EXAMPLE · DEPENDENT ← HEAD

三 ← 匹

‘Keeps’ (飼う) stands in for ‘has’ in the sense of owning a pet.

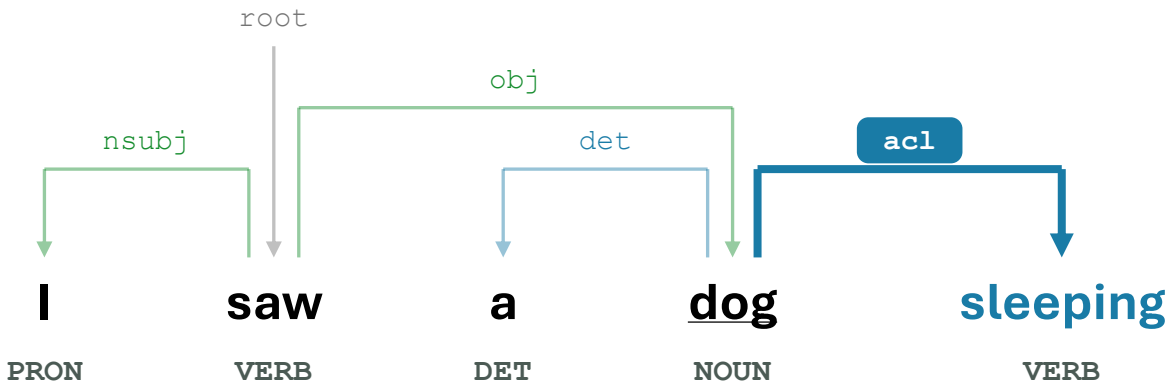
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

`acl` Clausal modifier of noun

A clause that modifies a noun, including relative clauses.



EXAMPLE · DEPENDENT ← HEAD

sleeping ← dog

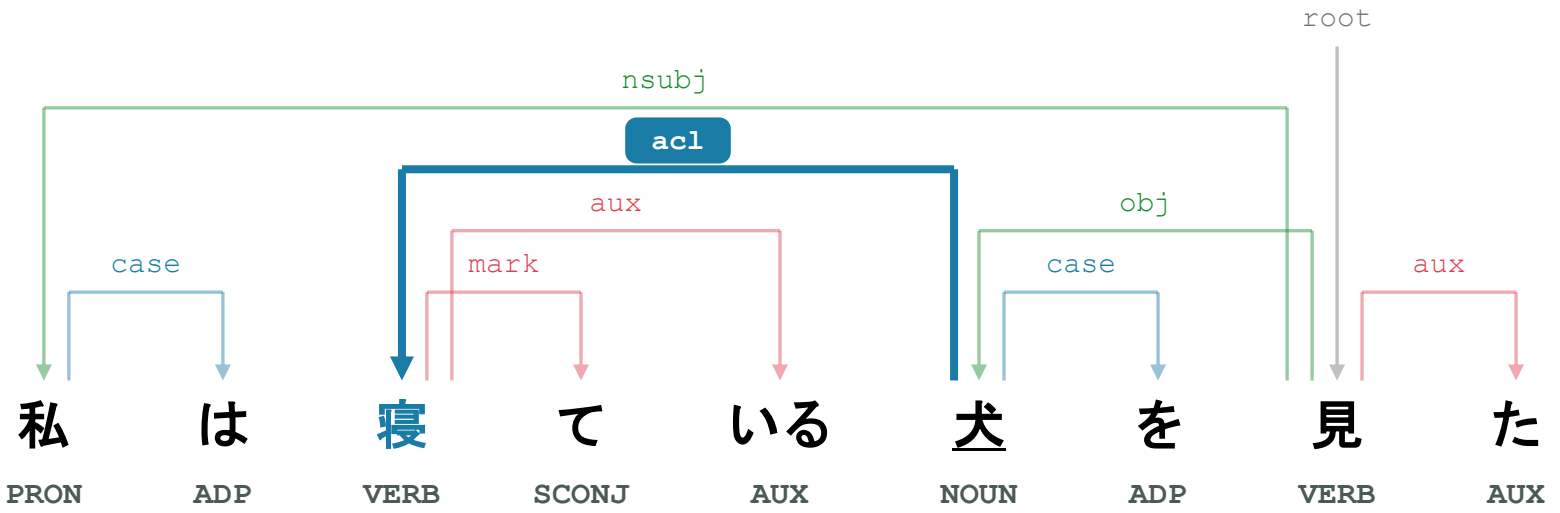
“I saw a dog sleeping”

NOMINAL DEPENDENTS

`nmod` `appos` `nummod` `acl` `amod` `det` `clf` `case`

acl Clausal modifier of noun

Relative clauses precede the noun they modify and need no relative pronoun.



‘I saw a dog sleeping.’

EXAMPLE · DEPENDENT ← HEAD

寝 ← 犬

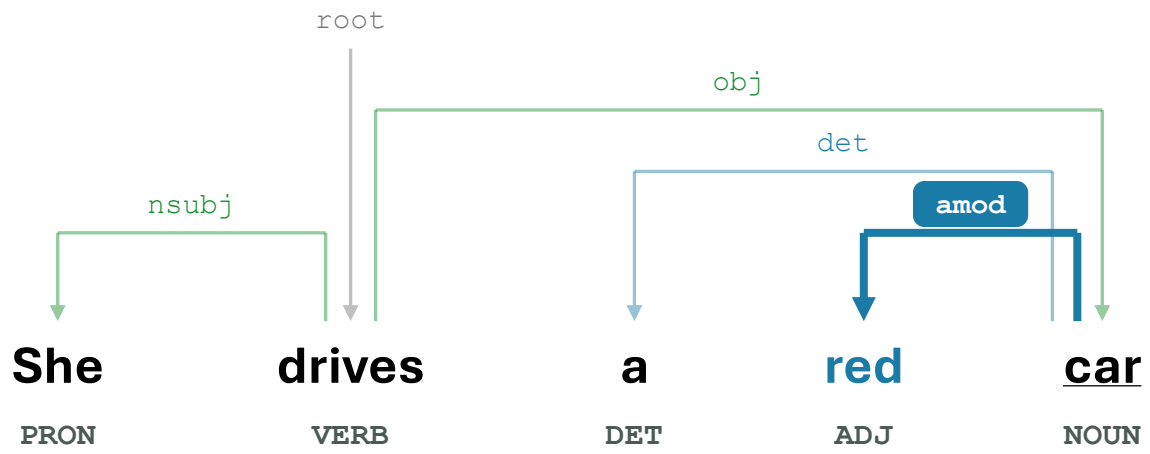
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

amod Adjectival modifier

An adjective that modifies a noun.



EXAMPLE · DEPENDENT ← HEAD

red ← car

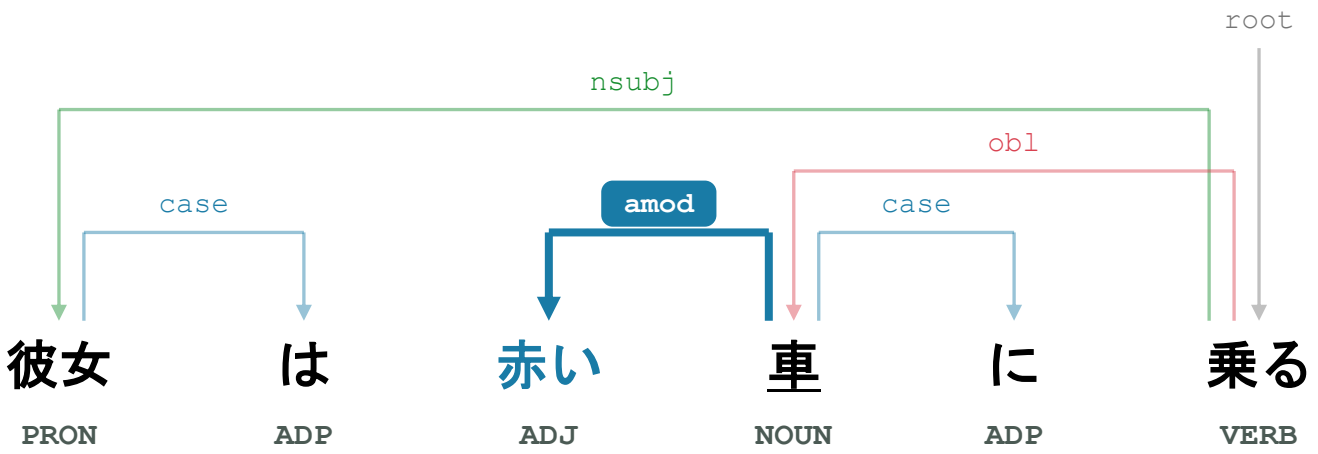
“She drives a red car”

NOMINAL DEPENDENTS

nmod appos nummod acl amod det clf case

amod Adjectival modifier

Adnominal adjectives precede the noun; UD Japanese treebanks usually analyse them as acl.



‘She rides a red car.’

EXAMPLE · DEPENDENT ← HEAD

赤い ← 車

‘Drives’ is 運転する, a light-verb compound, so ‘rides’ (乗る) is used.

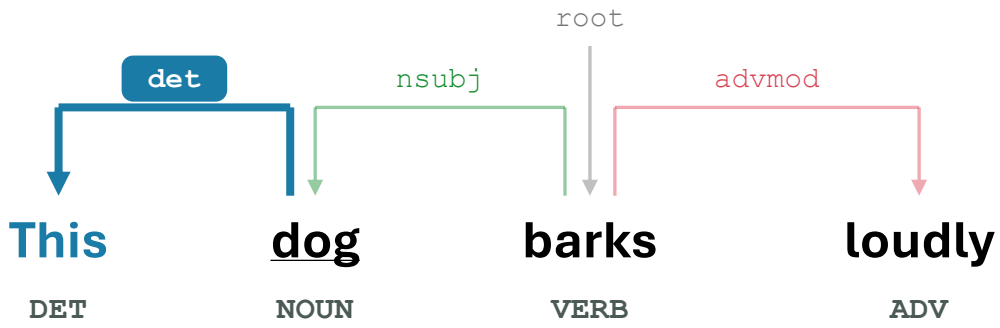
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

det Determiner

A word that specifies a noun: the, a, this, some, every.



EXAMPLE · DEPENDENT ← HEAD

This ← dog

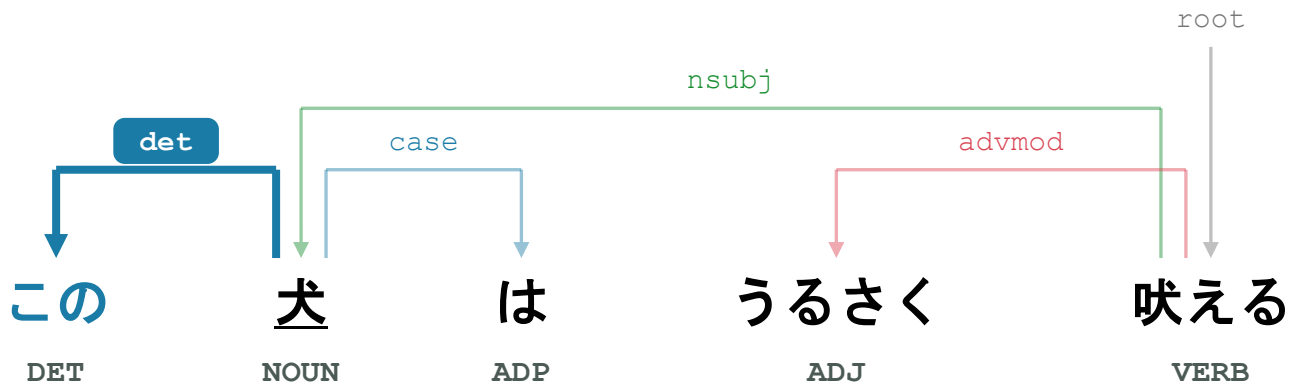
“This dog barks loudly”

NOMINAL DEPENDENTS

nmod appos nummod acl amod **det** clf case

det Determiner

A closed set of adnominals such as この, その, あの and ある modifies the noun.



‘This dog barks loudly.’

EXAMPLE · DEPENDENT ← HEAD

この ← 犬

‘Loudly’ is rendered as うるさく (‘noisily’).

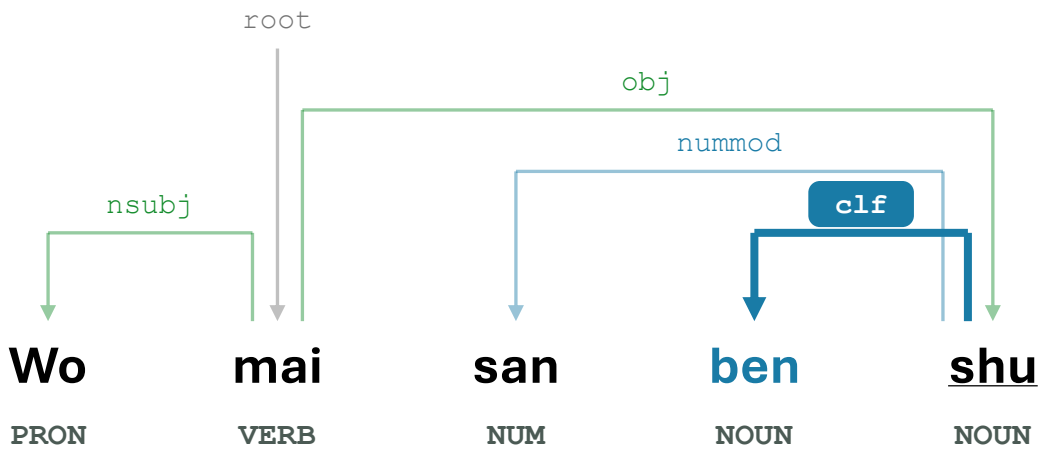
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

clf Classifier

A counting-unit word between number and noun (Mandarin, Thai…).



EXAMPLE · DEPENDENT ← HEAD

ben ← shu

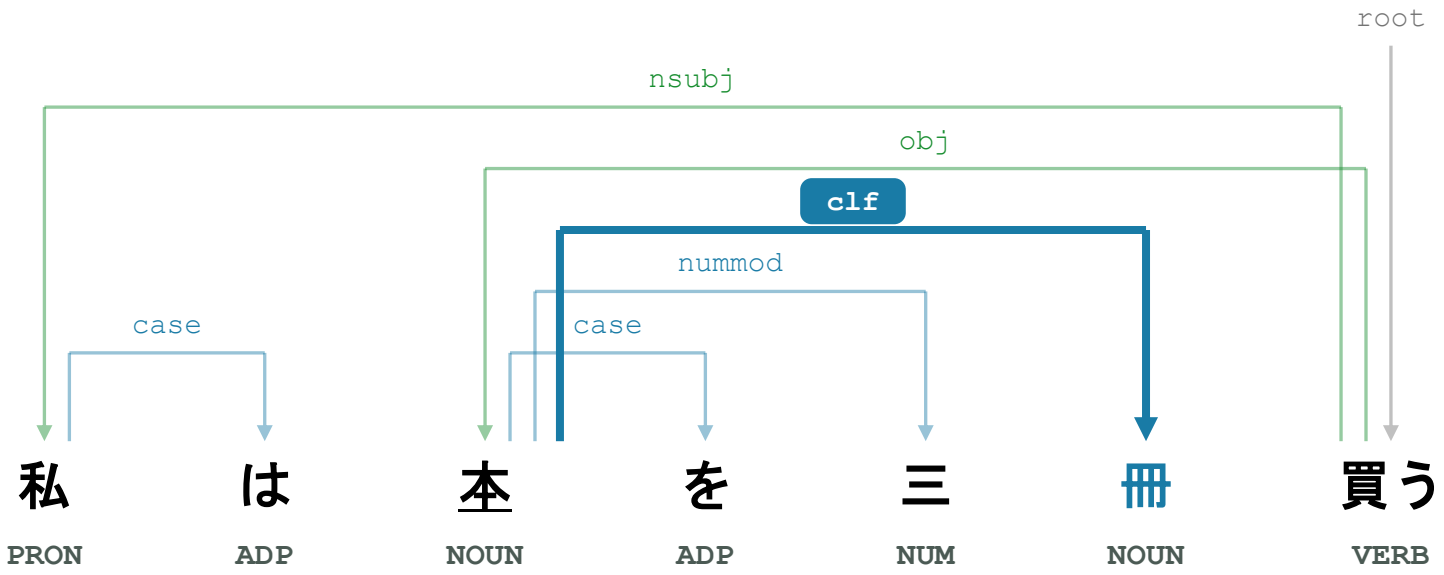
“Wo mai san ben shu” Mandarin (pinyin): ‘I buy three (volumes of) books’.

NOMINAL DEPENDENTS

nmod · appos · nummod · acl · amod · det · **clf** · case

clf Classifier

Japanese counters follow numerals; UD Japanese treebanks do not annotate them as clf.



‘I buy three books.’

EXAMPLE · DEPENDENT ← HEAD

冊 ← 本

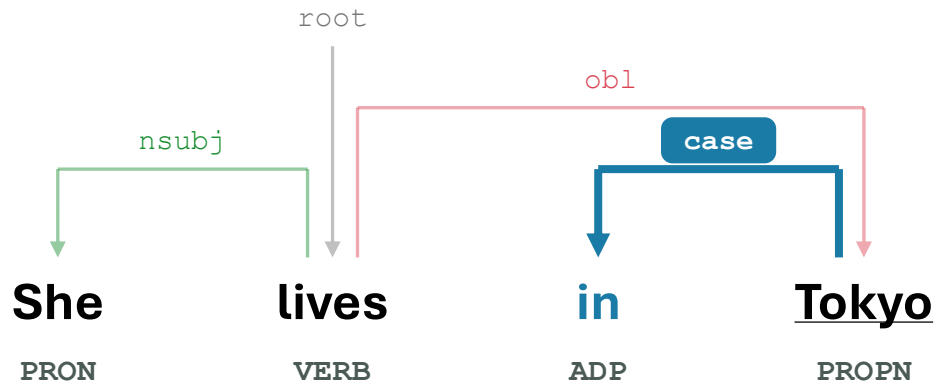
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

case Case marking

A preposition or case marker attached to its noun.



EXAMPLE · DEPENDENT ← HEAD

in ← Tokyo

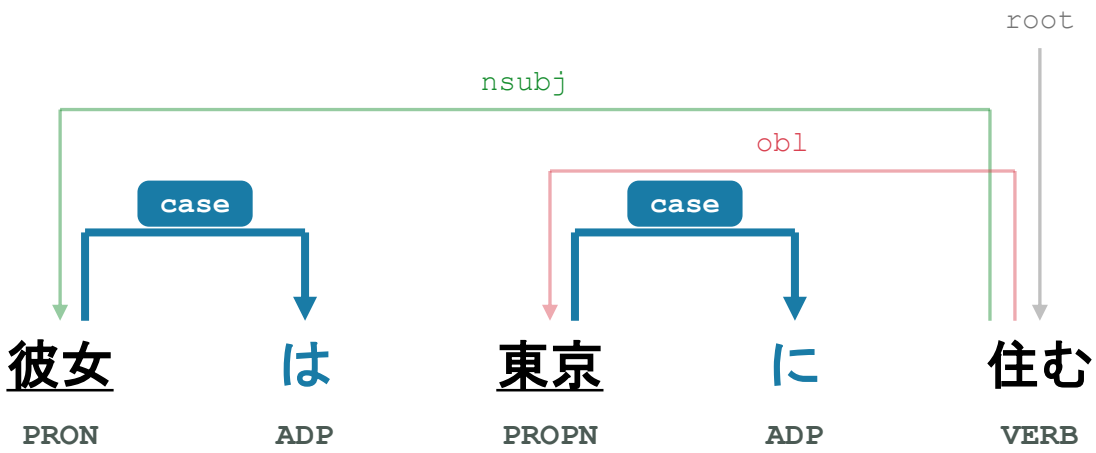
“She lives in Tokyo”

NOMINAL DEPENDENTS

nmod appos nummod acl amod det clf **case**

case Case marking

Postpositional case particles (が, を, に, で, の, は) follow their noun.



‘She lives in Tokyo.’

EXAMPLE · DEPENDENT ← HEAD

は ← 彼女　に ← 東京

UD JAPANESE

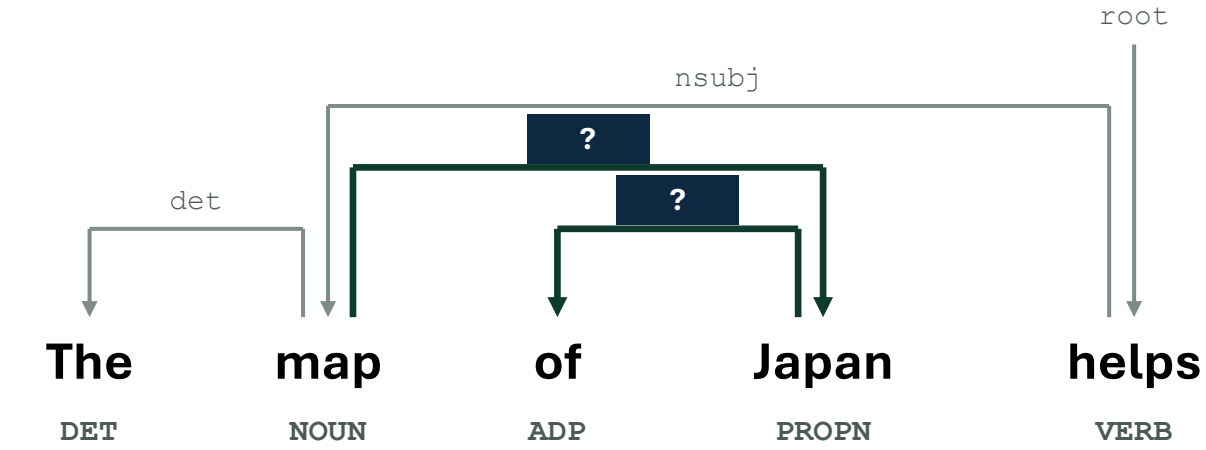
Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

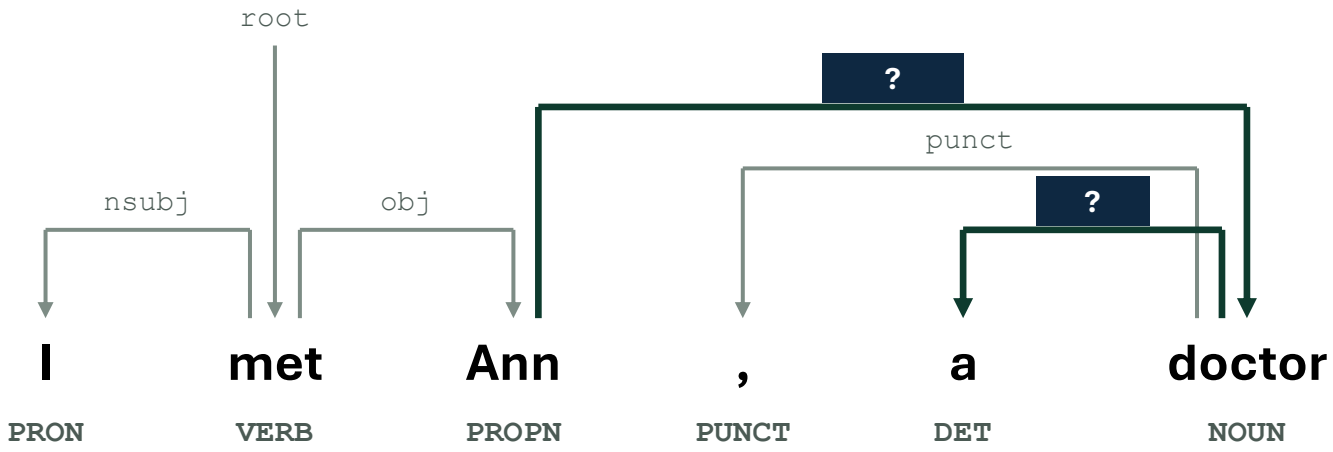
Practice 3 (1/2)

Noun Dependents · Click to reveal the hidden relation labels

1



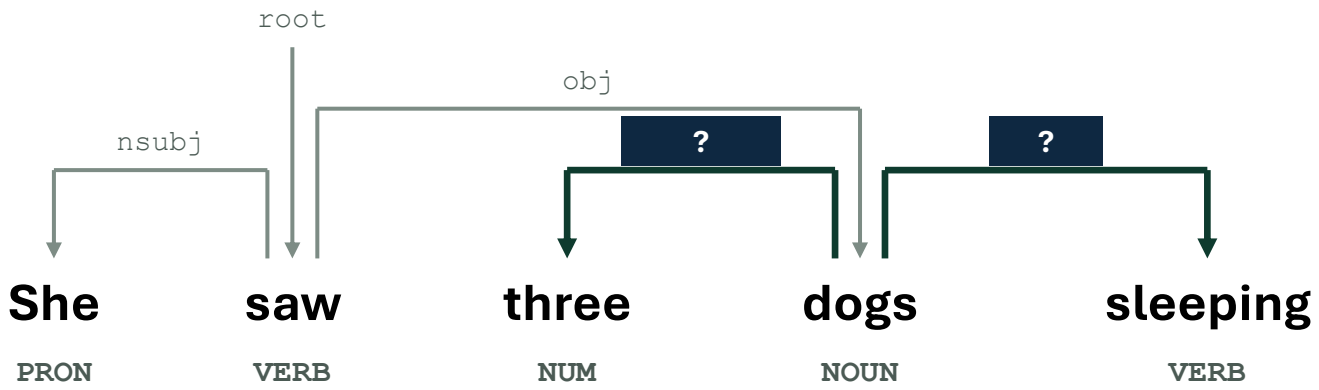
2



Practice 3 (2/2)

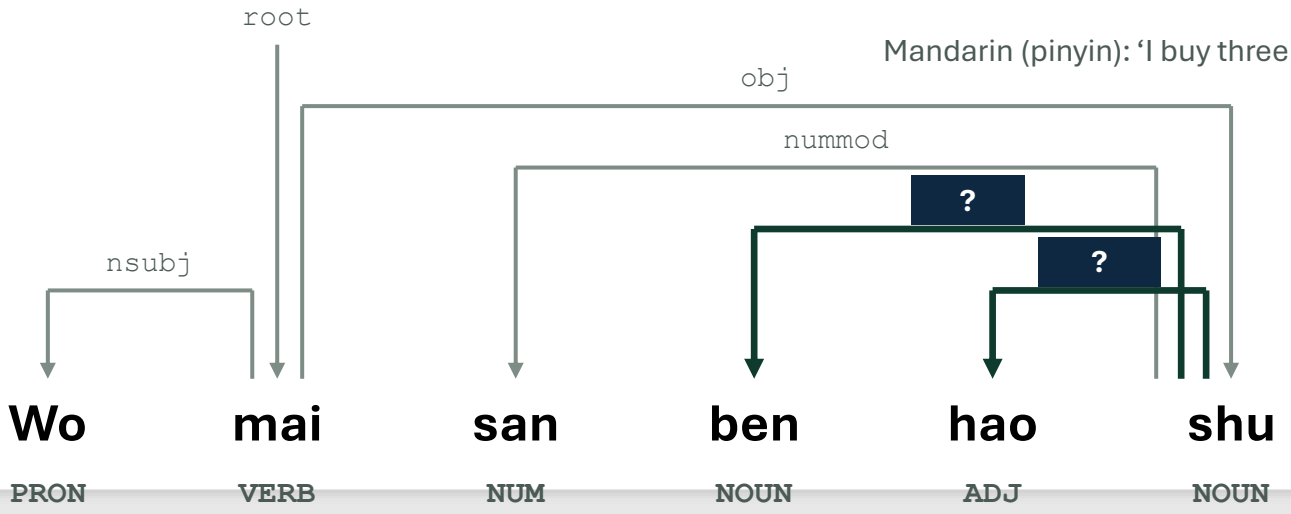
Noun Dependents · Click to reveal the hidden relation labels

3



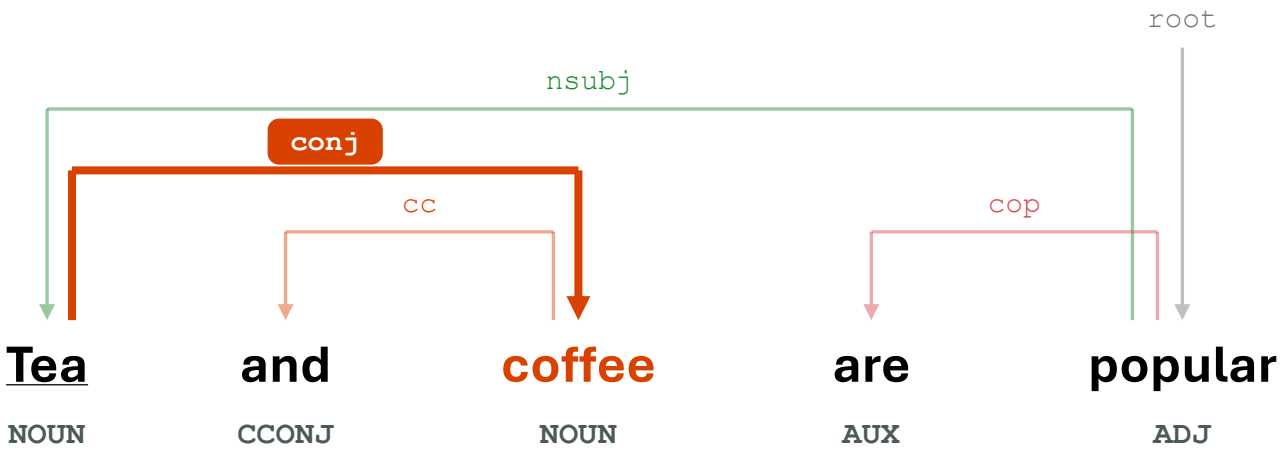
4

Mandarin (pinyin): ‘I buy three good books’.



conj Conjunct

A coordinated item attached to the first item of the group.



EXAMPLE · DEPENDENT ← HEAD

coffee ← Tea

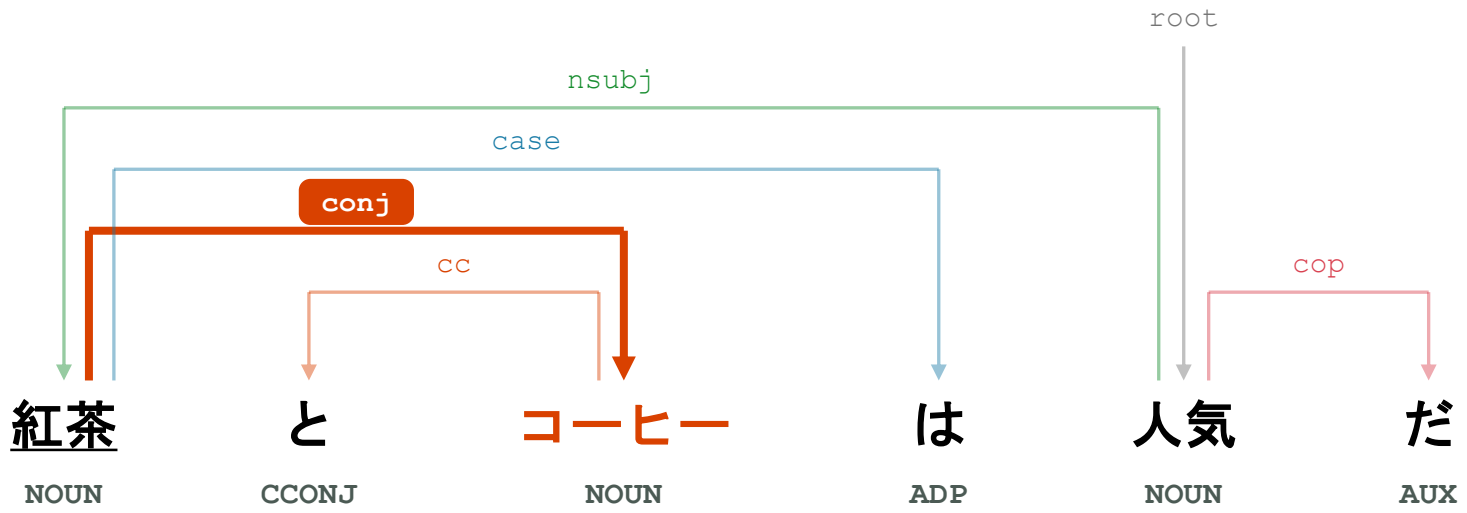
“Tea and coffee are popular”

COORDINATION

conj cc

conj Conjunct

Coordinated nouns are linked by と or や, with the first noun as the head.



'Tea and coffee are popular.'

EXAMPLE · DEPENDENT ← HEAD

コーヒー ← 紅茶

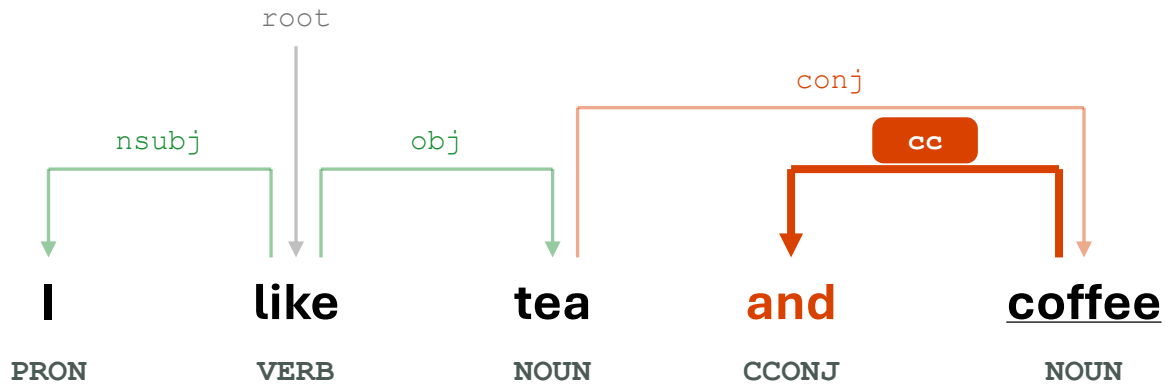
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

CC Coordinating conjunction

A conjunction linking conjuncts: and, or, but.



EXAMPLE · DEPENDENT ← HEAD

and ← coffee

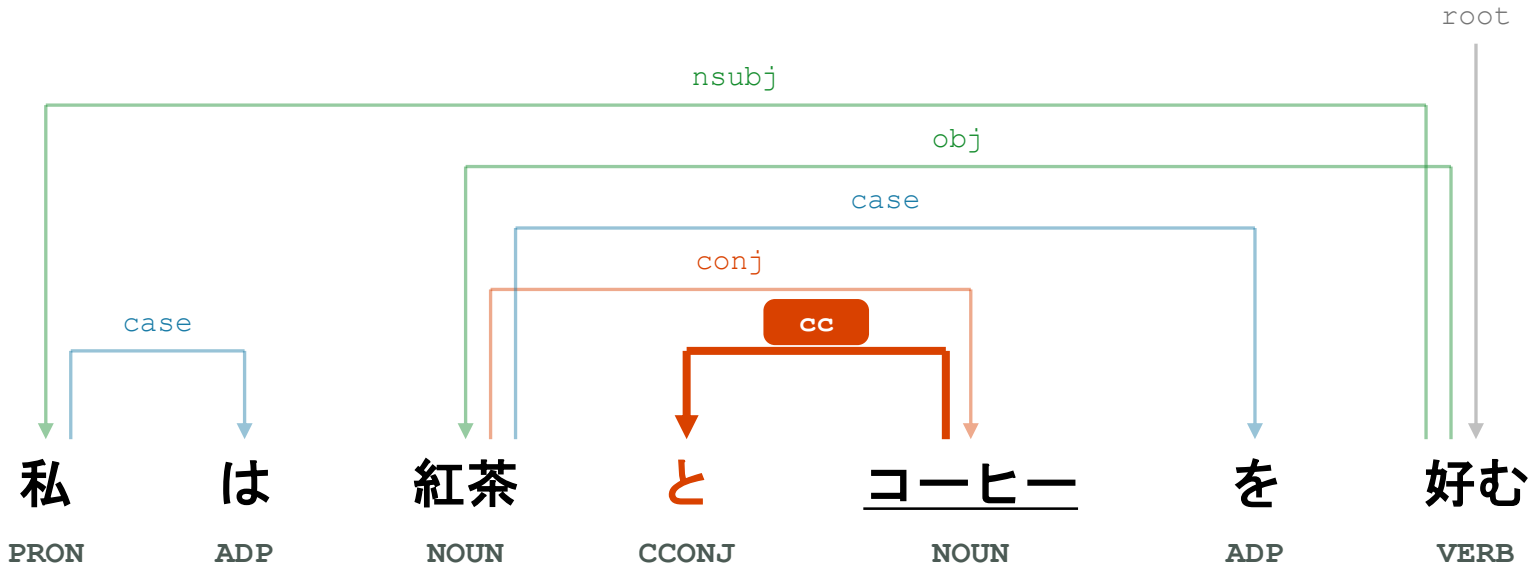
“I like tea and coffee”

COORDINATION

conj cc

CC Coordinating conjunction

と and や link conjuncts; しかし and そして link sentences.



‘I like tea and coffee.’

EXAMPLE · DEPENDENT ← HEAD

と ← コーヒー

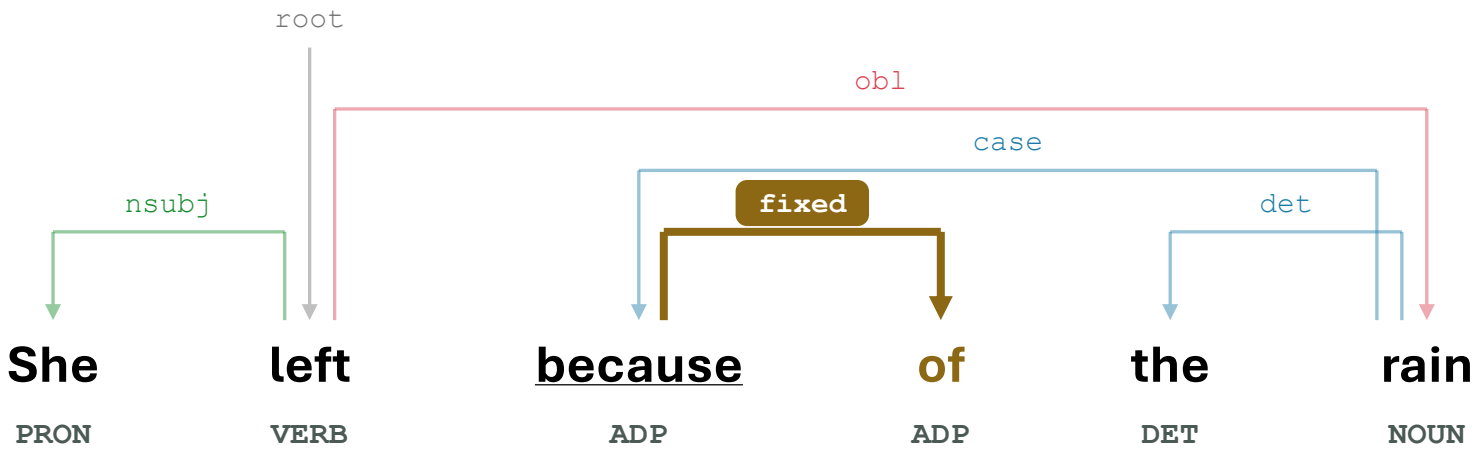
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

`fixed` Fixed multiword expression

Part of a fixed multiword function expression, like ‘because of’.



EXAMPLE · DEPENDENT ← HEAD

of ← because

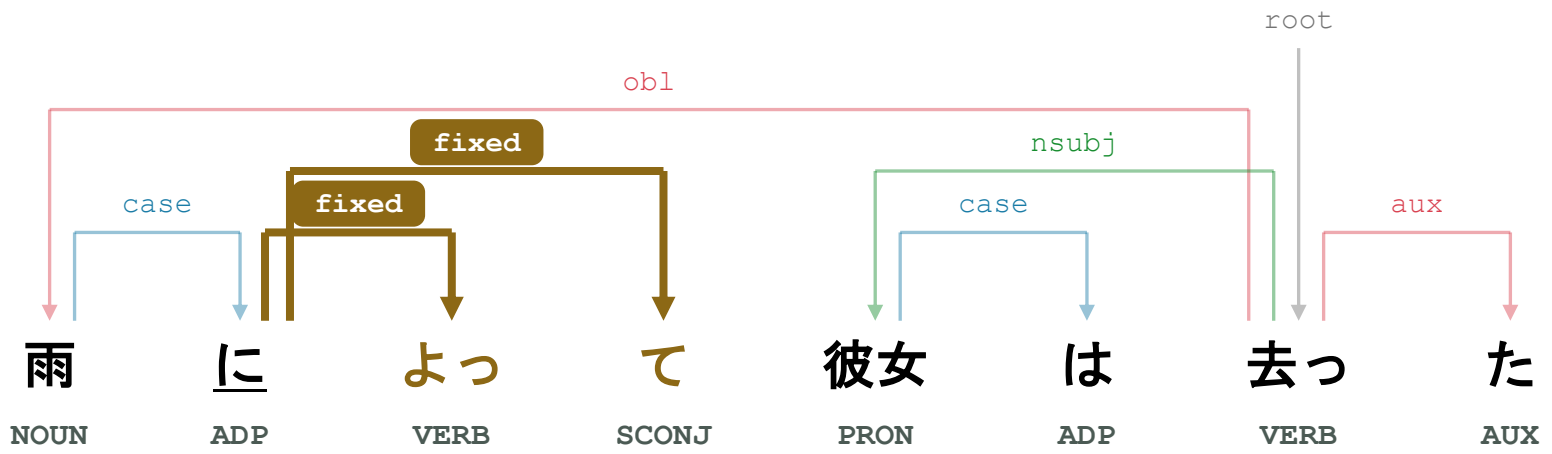
“She left because of the rain”

HEADLESS

`fixed` `flat`

fixed Fixed multiword expression

Multiword function expressions such as によって work like a single postposition.



‘Because of the rain, she left.’

EXAMPLE · DEPENDENT ← HEAD

よっ ← に　て ← に

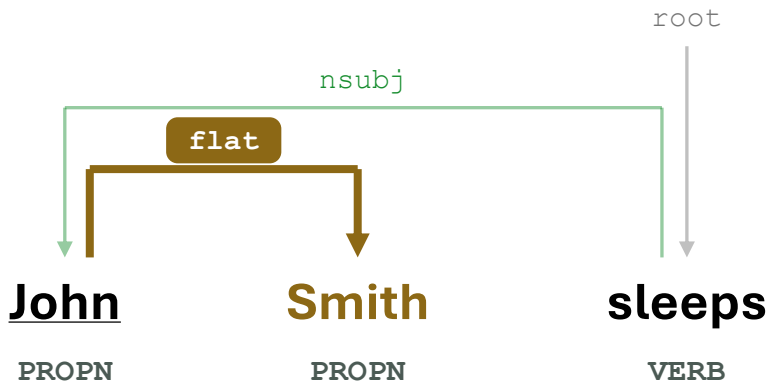
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

`flat` Flat expression

Parts of a name or foreign phrase, with no internal head.



EXAMPLE · DEPENDENT ← HEAD

Smith ← John

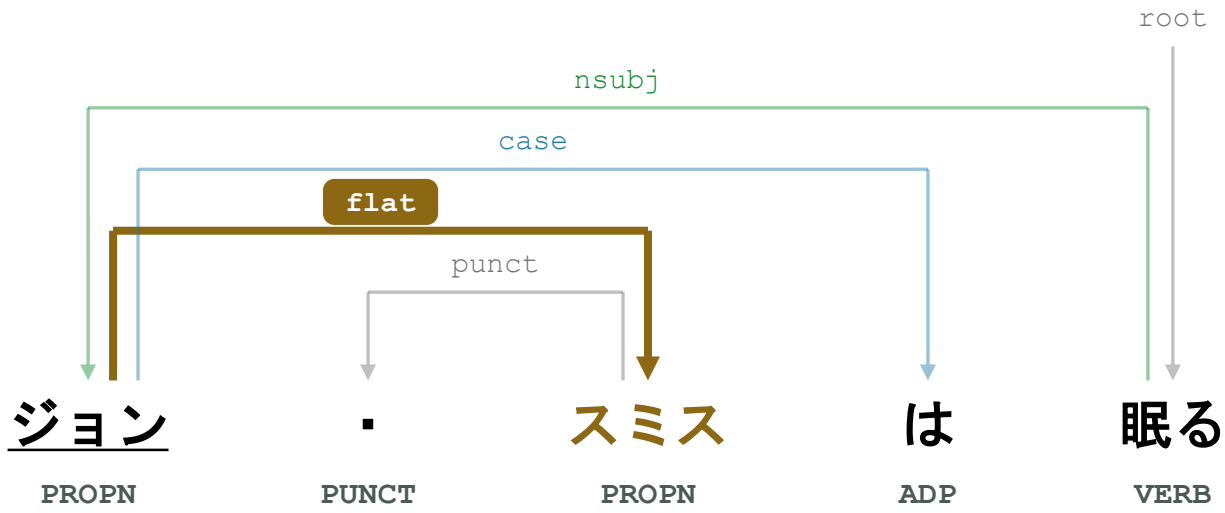
“John Smith sleeps”

HEADLESS

`fixed` `flat`

flat Flat expression

UD Japanese usually analyses personal names as compound; here flat is used as defined in UD.



‘John Smith sleeps.’

EXAMPLE · DEPENDENT ← HEAD

スミス ← ジョン

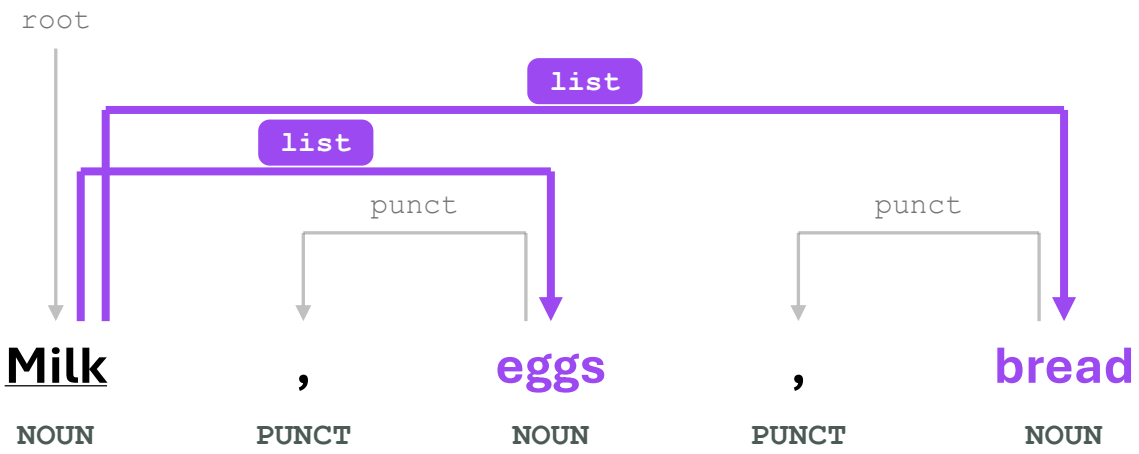
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

`list` List

Items of a list, each attached to the first item.



EXAMPLE · DEPENDENT ← HEAD

eggs ← Milk **bread ← Milk**

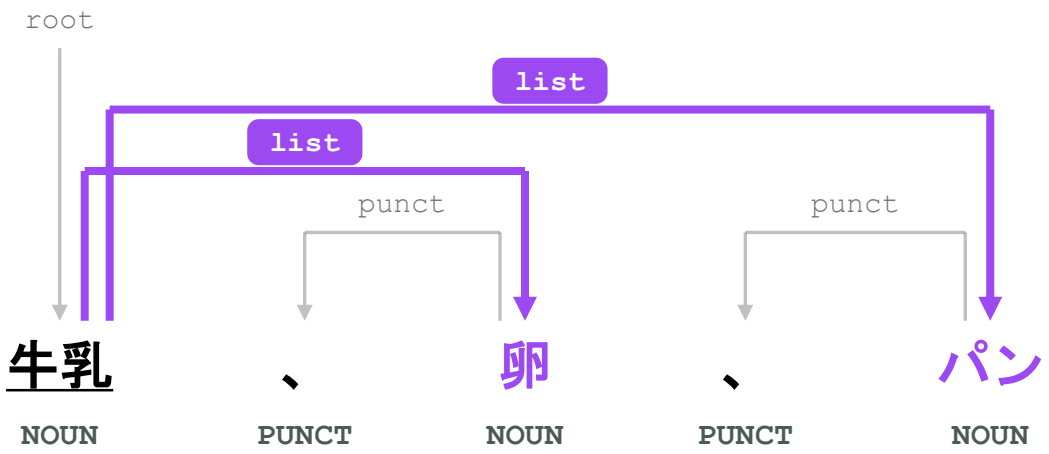
“Milk, eggs, bread”

LOOSE

`list` `parataxis`

list List

Enumerations separated by 、 can be treated as a list under the universal definition.



‘Milk, eggs, bread.’

EXAMPLE · DEPENDENT ← HEAD

卵 ← 牛乳　パン ← 牛乳

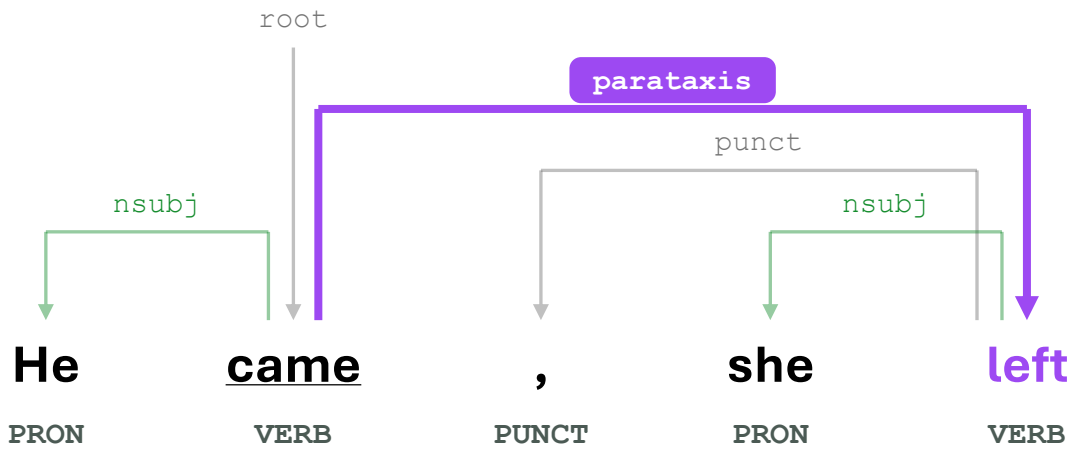
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

parataxis Parataxis

Clauses placed side by side without a linking word.



EXAMPLE · DEPENDENT ← HEAD

left ← came

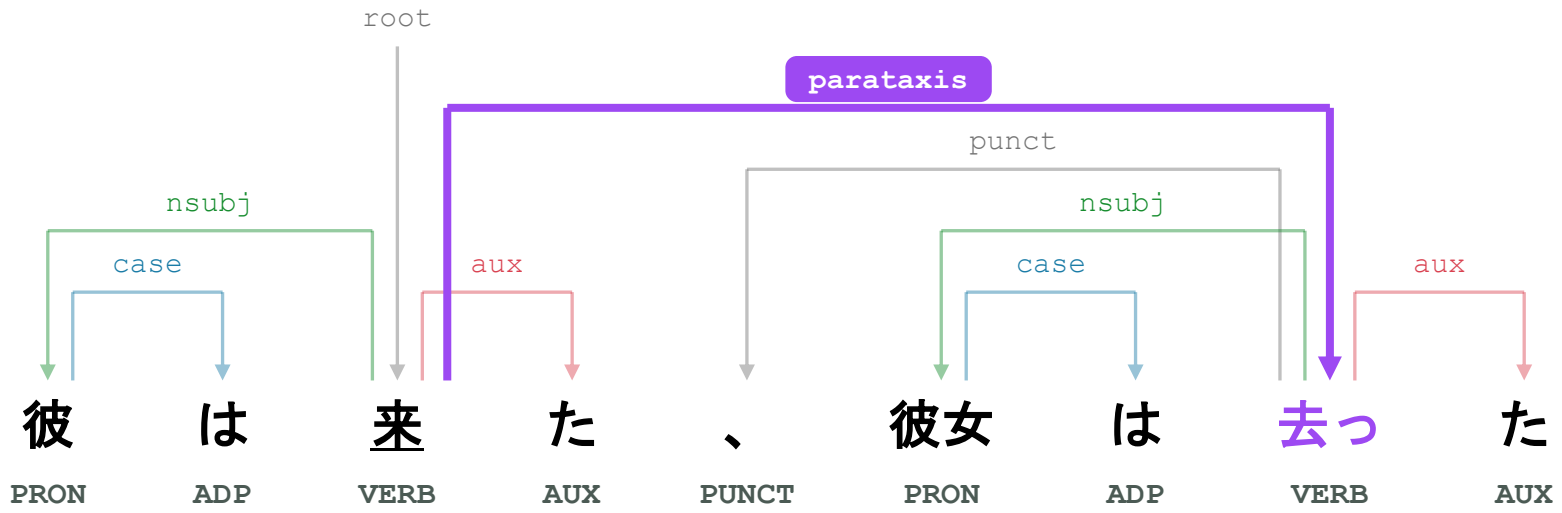
“He came, she left”

LOOSE

list parataxis

parataxis Parataxis

Clauses can be juxtaposed with a comma, without any conjunctive particle.



‘He came, she left.’

EXAMPLE · DEPENDENT ← HEAD

去っ ← 来

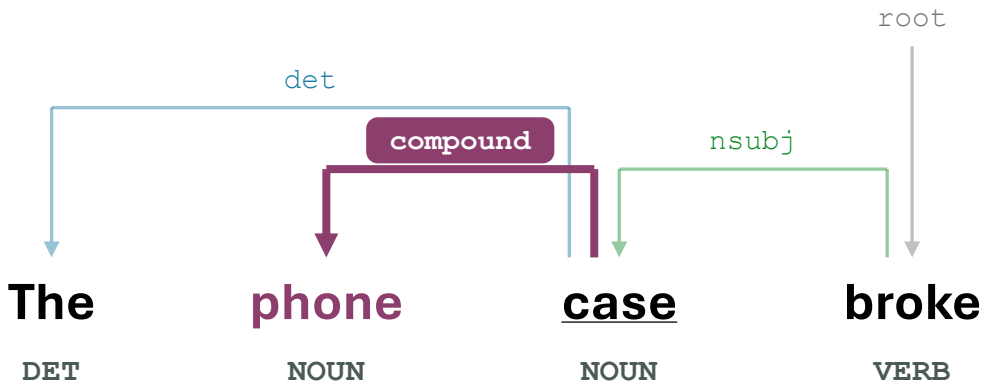
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

compound Compound

A word forming a compound with its head noun: phone case.



EXAMPLE · DEPENDENT ← HEAD

phone ← case

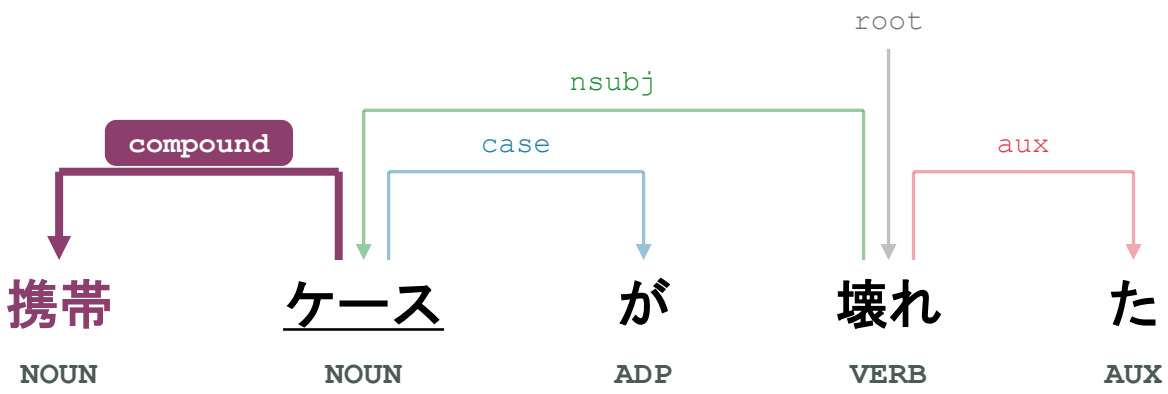
“The phone case broke”

SPECIAL

compound orphan goeswith reparandum

compound Compound

Noun compounds are common; the first noun modifies the second, written without a space.



‘The phone case broke.’

EXAMPLE · DEPENDENT ← HEAD

携帯 ← ケース

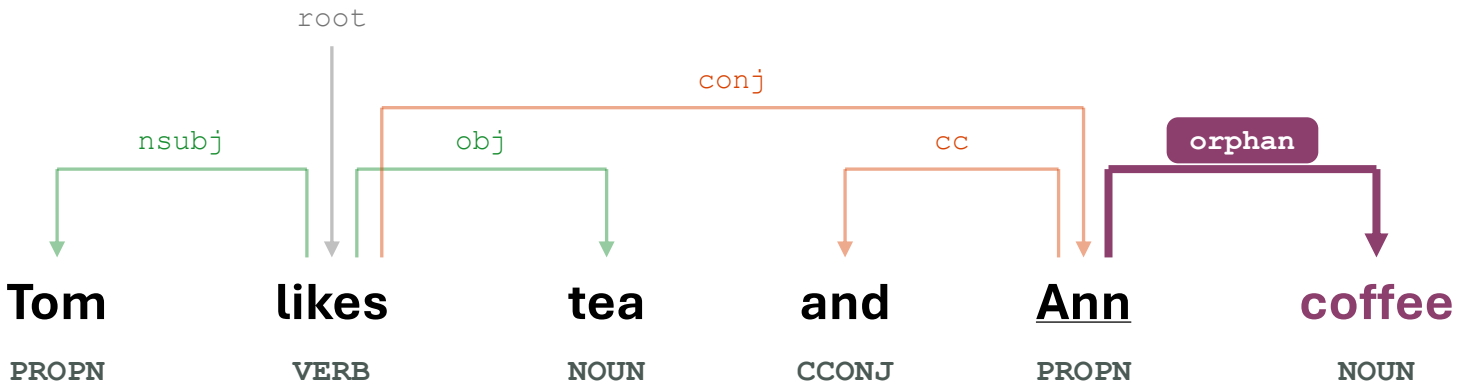
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

orphan Orphan

A leftover item promoted to head when a word is omitted (ellipsis).



EXAMPLE · DEPENDENT ← HEAD

coffee ← Ann

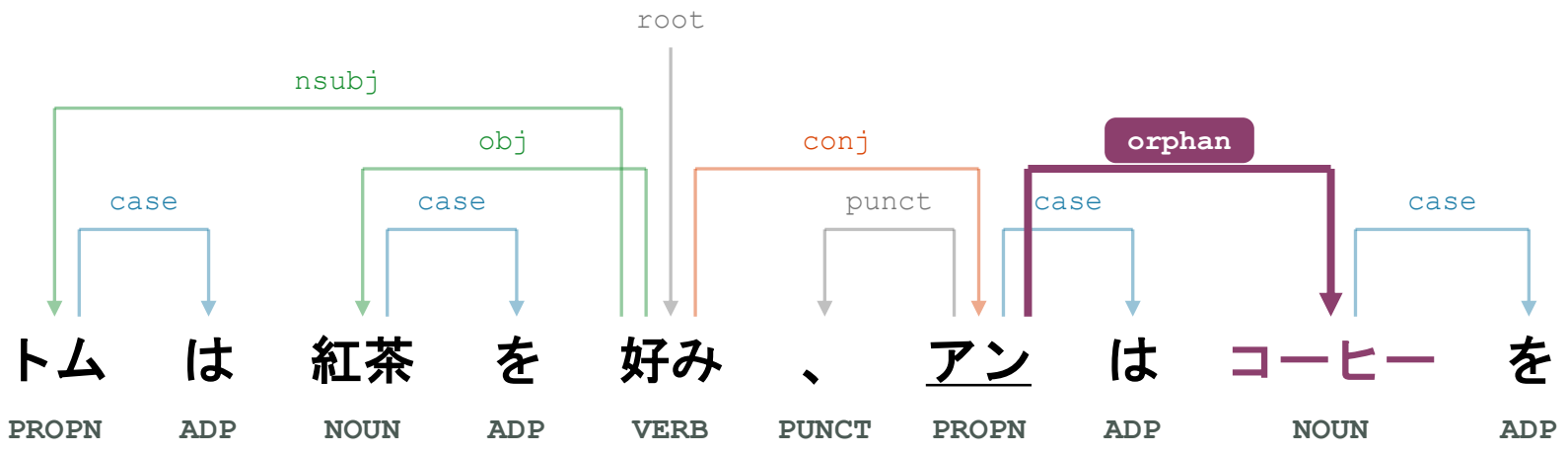
“Tom likes tea and Ann coffee”

SPECIAL

compound | orphan | goeswith | reparandum

orphan Orphan

When the verb of the second clause is omitted, its remaining arguments become orphans.



‘Tom likes tea, and Ann coffee.’

EXAMPLE · DEPENDENT ← HEAD

コーヒー ← アン

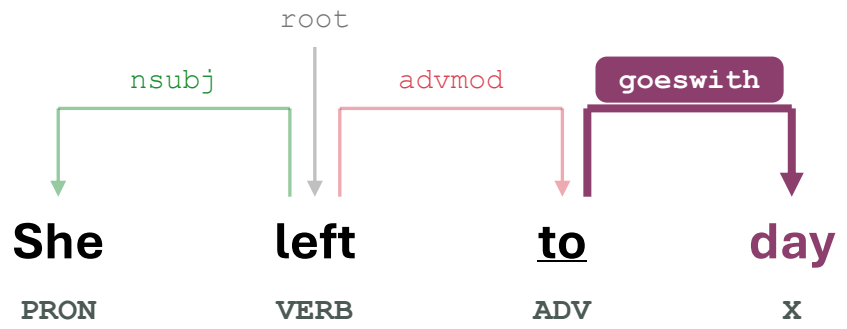
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

goeswith Goes with

The second part of a word wrongly split in the text.



EXAMPLE · DEPENDENT ← HEAD

day ← to

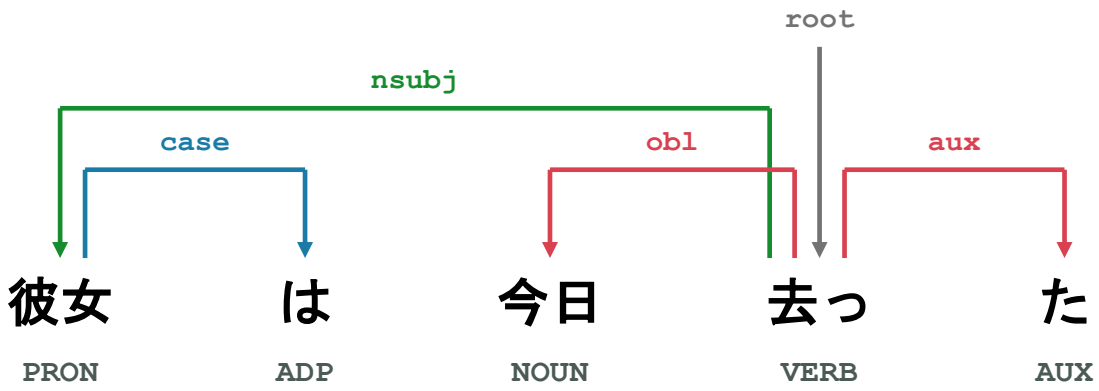
“She left to day” ‘to day’ is how the writer split ‘today’.

SPECIAL

compound orphan goeswith reparandum

Column Split words across languages

English can split one word by mistake (‘to day’); Japanese has no spaces, so goeswith is not needed.



‘She left today.’

ENGLISH vs JAPANESE

English: day ← to (goeswith)

Japanese: 今日 is a single token

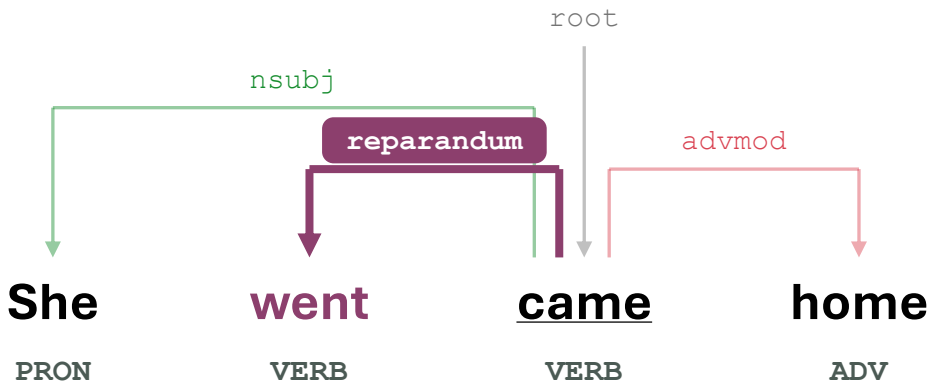
UD JAPANESE

Not used in Japanese: there is no spacing between words to mis-split.

Hand-annotated after UD Japanese conventions.

`reparandum` Overridden disfluency

A word the speaker abandons and corrects (speech repair).



EXAMPLE · DEPENDENT ← HEAD

went ← came

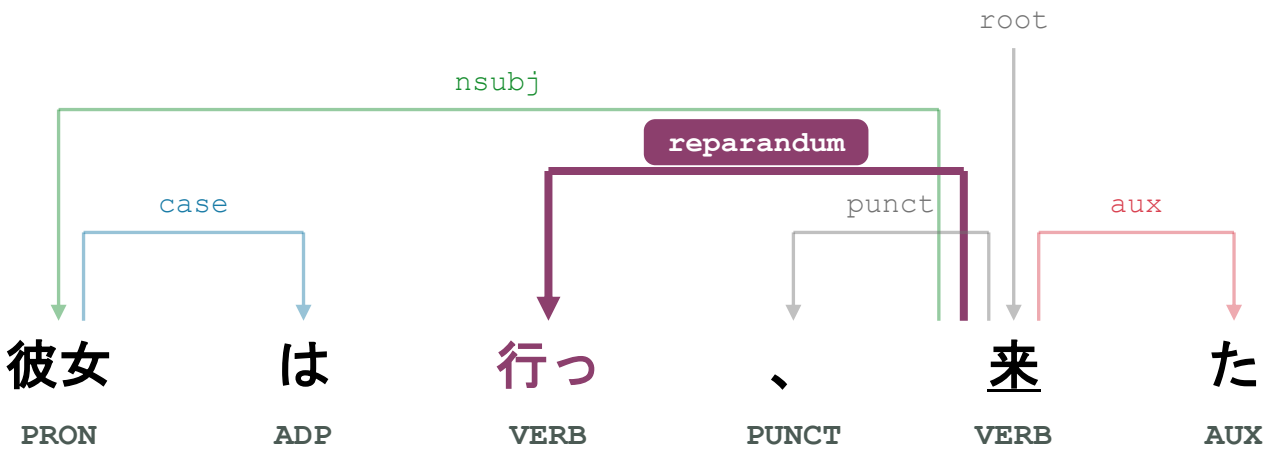
“She went came home” The speaker corrects ‘went’ to ‘came’.

SPECIAL

`compound` `orphan` `goeswith` `reparandum`

reparandum Overridden disfluency

In spoken transcripts, the abandoned word before a self-correction is marked as reparandum.



‘She went— came.’

EXAMPLE · DEPENDENT ← HEAD

行っ ← 来

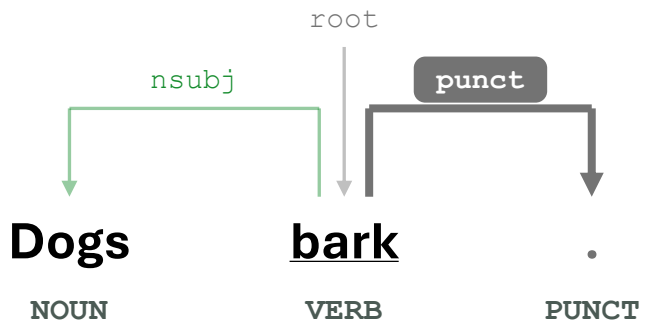
UD JAPANESE

Not in the UD Japanese-GSD relation list; the example applies the universal definition.

Hand-annotated after UD Japanese conventions.

punct Punctuation

Punctuation marks, attached to the head of their phrase.



EXAMPLE · DEPENDENT ← HEAD

. ← bark

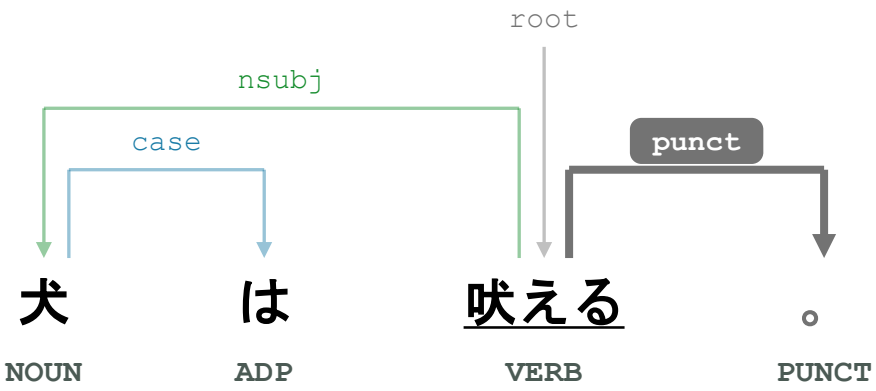
“Dogs bark.”

OTHER

punct root dep

punct Punctuation

、 and 。 attach to the head of the phrase or clause they close.



‘Dogs bark.’

EXAMPLE · DEPENDENT ← HEAD

。 ← 吠える

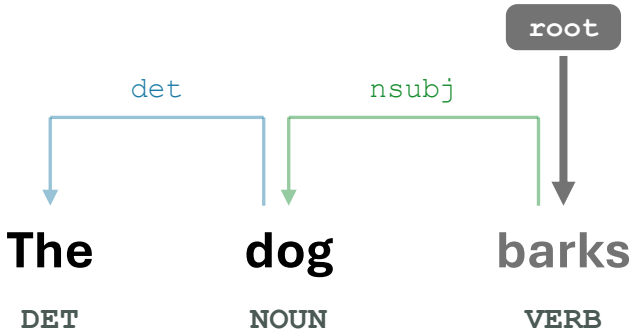
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

root Root

The word with no head: the entry point of the whole tree.



EXAMPLE · DEPENDENT ← HEAD

barks ← (root)

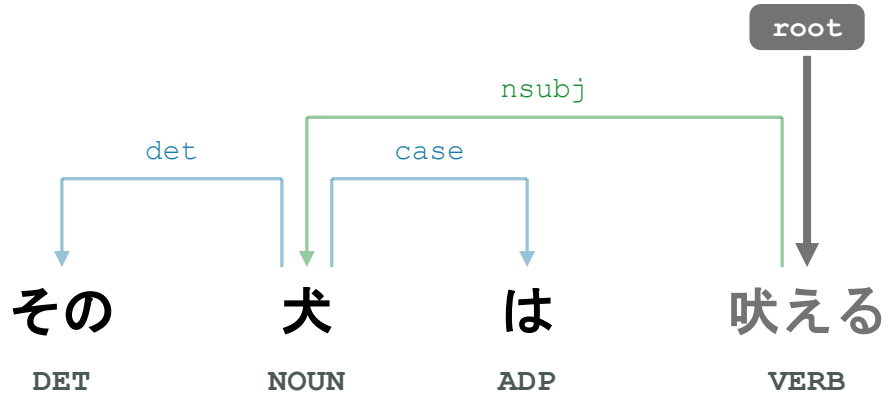
“The dog barks”

OTHER

punct root dep

root Root

The sentence-final predicate is the root, since Japanese is head-final.



‘The dog barks.’

EXAMPLE · DEPENDENT ← HEAD

吠える ← (root)

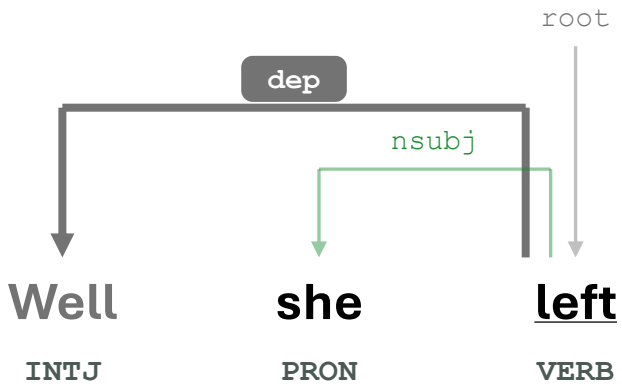
UD JAPANESE

Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

dep Unspecified dependency

A fallback when no more precise relation can be determined.



EXAMPLE · DEPENDENT ← HEAD

Well ← left

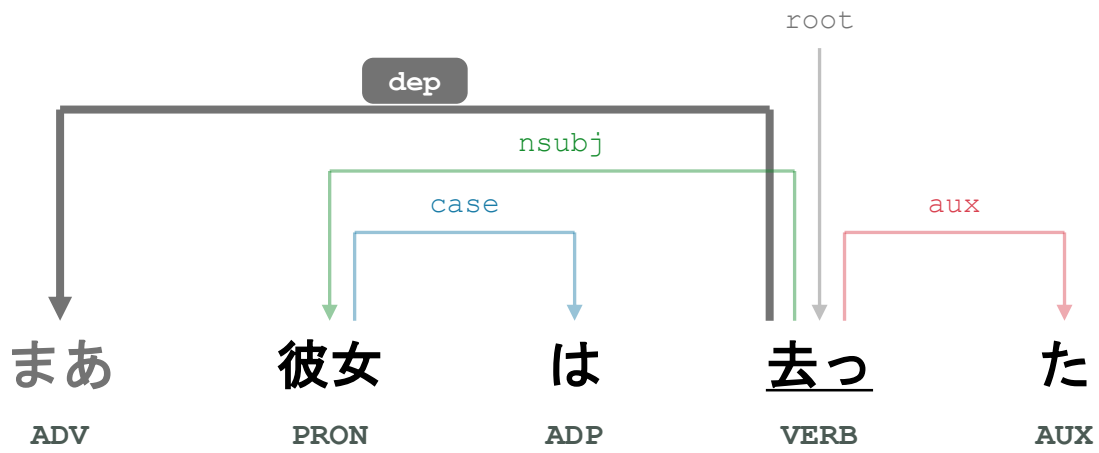
“Well she left” Illustrative: ‘Well’ could not be classified more precisely.

OTHER

punct root dep

dep Unspecified dependency

A fallback label when no more specific relation can be assigned.



‘Well, she left.’

EXAMPLE · DEPENDENT ← HEAD

まあ ← 去っ

Illustrative.

UD JAPANESE

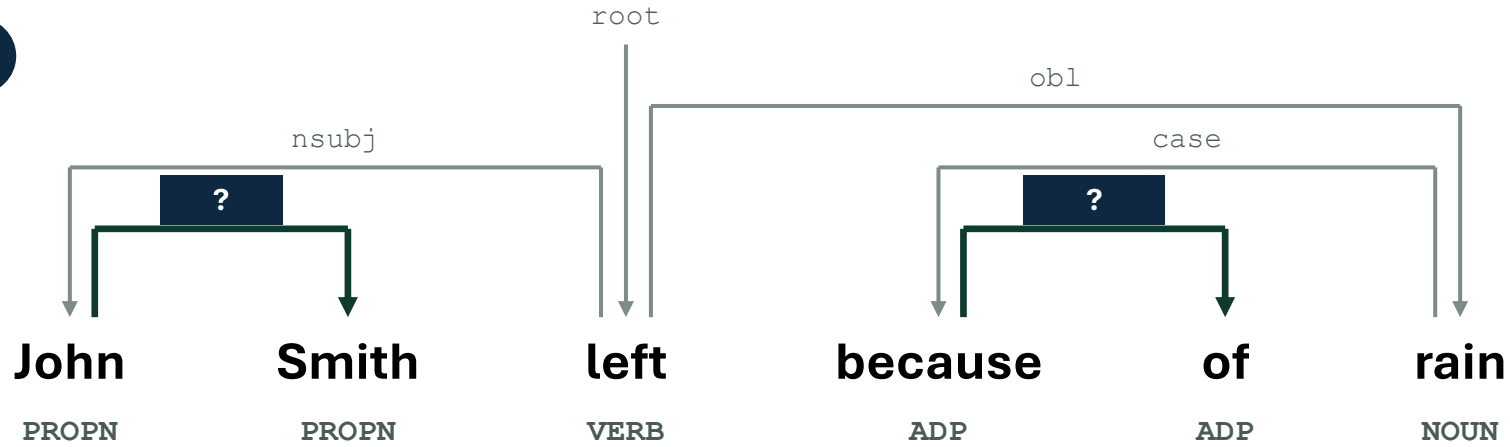
Attested in UD Japanese-GSD.

Hand-annotated after UD Japanese conventions.

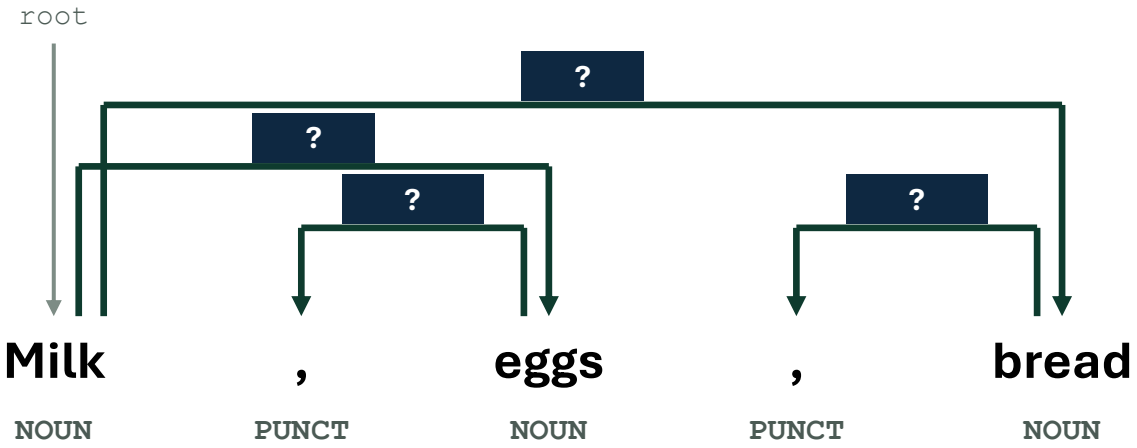
Practice 4 (1/4)

Other Relations · Click to reveal the hidden relation labels

1



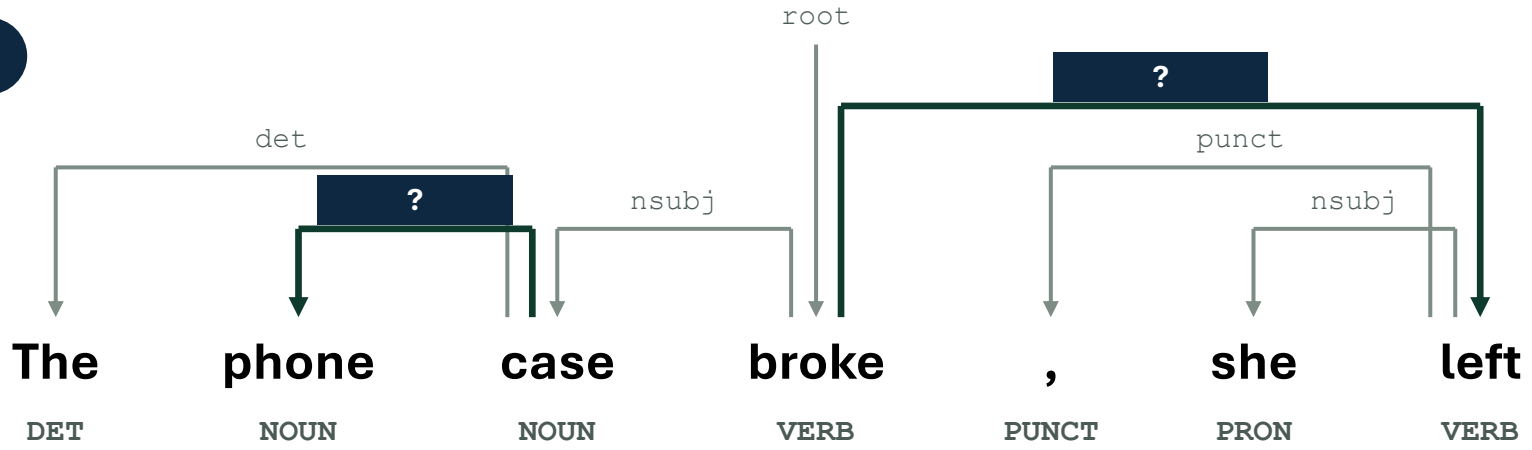
2



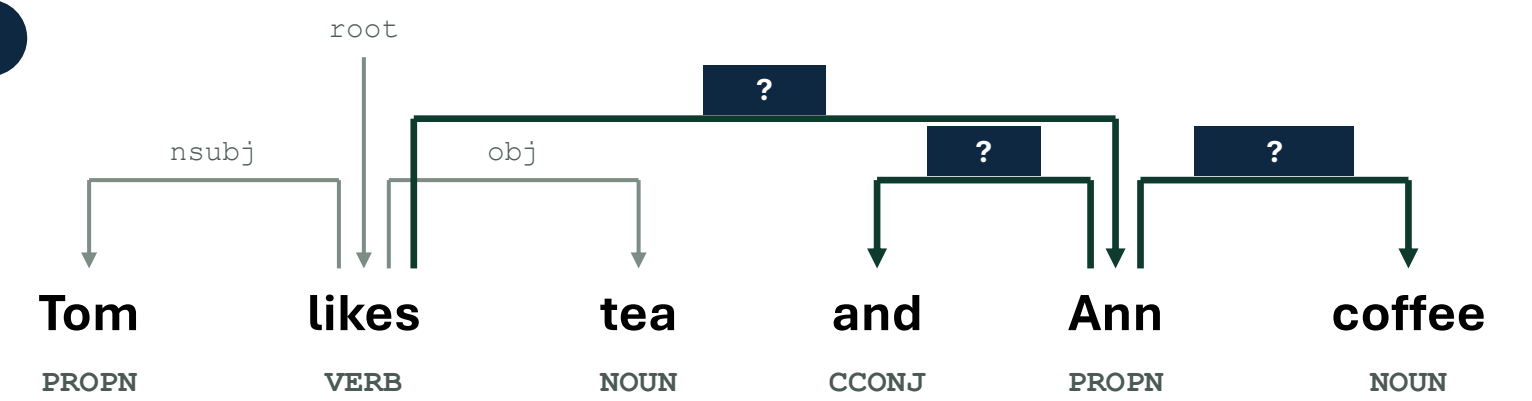
Practice 4 (2/4)

Other Relations · Click to reveal the hidden relation labels

3



4

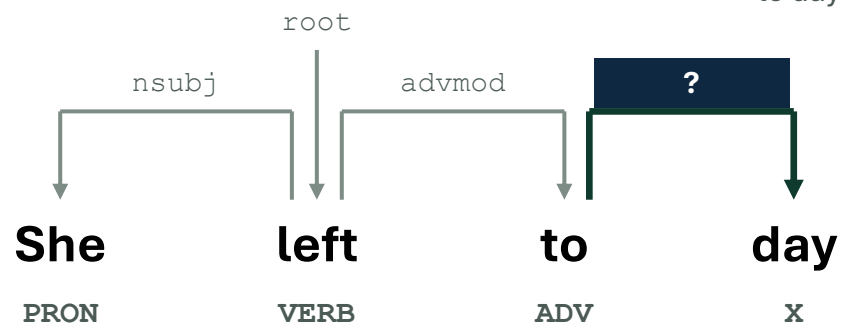


Practice 4 (3/4)

Other Relations · Click to reveal the hidden relation labels

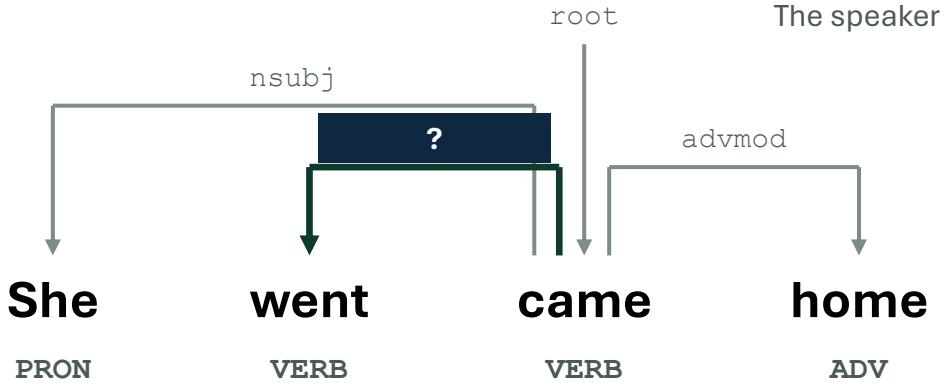
5

‘to day’ is how the writer split ‘today’.



6

The speaker corrects ‘went’ to ‘came’.



Practice 4 (4/4)

Other Relations · Click to reveal the hidden relation labels

7

